Equation (4.45) for this case is

$$\theta(t) = \sum_{k=1}^{2} \xi^{(k)} A_k \cos \omega_k t ,$$

so upon writing out the components of $\theta(t)$, we get

$$\theta_1(t) = \frac{\alpha_1 m_1 + \alpha_2 m_2}{m_1 + m_2} \cos \omega_1 t + \frac{m_2(\alpha_1 - \alpha_2)}{m_1 + m_2} \cos \omega_2 t ,$$

$$\theta_2(t) = \frac{\alpha_1 m_1 + \alpha_2 m_2}{m_1 + m_2} \cos \omega_1 t - \frac{m_1(\alpha_1 - \alpha_2)}{m_1 + m_2} \cos \omega_2 t .$$

Note that if $\alpha_1 = \alpha_2 = \alpha$, then we have motion with only ω_1 present; if $\alpha_1 = (\mu/m_1)\alpha$, $\alpha_2 = -(\mu/m_2)\alpha$, then we have motion with only ω_2 present. These then are two sets of initial conditions which excite the two possible normal modes.

4.11 PERTURBATION THEORY—NONDEGENERATE CASE

Perturbation theory is a very useful technique by which one can obtain approximate solutions to eigenvalue-eigenvector problems that are too complicated to be solved exactly. The following situation is a common one. We want to find the eigenvalues and eigenvectors of a self-adjoint operator A which may be split up into the sum of two self-adjoint operators:

$$A = A_0 + \epsilon A_1 . \tag{4.49}$$

We suppose that we know something about A_0. For example, we might know some eigenvalue and its associated eigenvector. We would like to find the corresponding eigenvalue and eigenvector of A, given that the influence of A_0 is known to be predominant. We have already anticipated this situation by writing the "perturbing" operator in the form ϵA_1, where ϵ is some small parameter.

Thus we assume that the equation

$$A_0 x_n^{(0)} = \lambda_n^{(0)} x_n^{(0)} \tag{4.50}$$

has been solved at least for some n. We now want to look for the corresponding solution of the original problem

$$A x_n = \lambda_n x_n \tag{4.51}$$

which has the form

$$\begin{aligned} \lambda_n &= \lambda_n^{(0)} + \epsilon \lambda_n^{(1)} + \cdots = \sum_{i=0}^{\infty} \epsilon^i \lambda_n^{(i)} , \\ x_n &= x_n^{(0)} + \epsilon x_n^{(1)} + \cdots = \sum_{i=0}^{\infty} \epsilon^i x_n^{(i)} . \end{aligned} \tag{4.52}$$

We shall not at this point discuss the convergence of these power series expansions in ϵ, except to observe that it would be astonishing if they converged for all ϵ.

By substituting the expansions into Eq. (4.51) and dropping for the moment the subscript n on the eigenvalues and eigenvectors, we obtain

$$(A_0 + \epsilon A_1) \sum_{i=0}^{\infty} \epsilon^i x^{(i)} = \left(\sum_{i=0}^{\infty} \epsilon^i \lambda^{(i)}\right)\left(\sum_{j=0}^{\infty} \epsilon^j x^{(j)}\right)$$

or

$$\sum_{i=0}^{\infty} \epsilon^i A_0 x^{(i)} + \sum_{i=0}^{\infty} \epsilon^{i+1} A_1 x^{(i)} = \sum_{i,j=0}^{\infty} \epsilon^{i+j} \lambda^{(i)} x^{(j)} . \tag{4.53}$$

To solve this, we equate coefficients of like powers of ϵ on the left- and right-hand sides of (4.53). The right-hand side is not in a good form for comparison, but if we reshuffle a little, we get

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \epsilon^{i+j} \lambda^{(i)} x^{(j)} = \sum_{\nu=0}^{\infty} \sum_{\mu=0}^{\nu} \epsilon^{\nu} \lambda^{(\mu)} x^{(\nu-\mu)} .$$

If we set $x^{(\alpha)} \equiv 0$ when $\alpha \leq -1$, then Eqs. (4.53) may be rewritten as

$$\sum_{\nu=0}^{\infty} \epsilon^{\nu} A_0 x^{(\nu)} + \sum_{\nu=0}^{\infty} \epsilon^{\nu} A_1 x^{(\nu-1)} = \sum_{\nu=0}^{\infty} \sum_{\mu=0}^{\nu} \epsilon^{\nu} \lambda^{(\mu)} x^{(\nu-\mu)} .$$

Now we equate powers of ϵ^{ν} to get

$$A_0 x^{(\nu)} + A_1 x^{(\nu-1)} = \sum_{\mu=0}^{\nu} \lambda^{(\mu)} x^{(\nu-\mu)} ,$$

or

$$[A_0 - \lambda^{(0)}] x^{(\nu)} = \sum_{\mu=1}^{\nu} \lambda^{(\mu)} x^{(\nu-\mu)} - A_1 x^{(\nu-1)} , \qquad \nu = 0, 1, \cdots \tag{4.54}$$

Here we write $[A - \lambda^{(0)} I]$ as $[A - \lambda^{(0)}]$ to simplify the notation. Thus we have an equation for $x^{(\nu)}$ in terms of $x^{(0)}, x^{(1)}, \cdots, x^{(\nu-1)}$, the lower-order functions. However, the right-hand side of the equation for $x^{(\nu)}$ also contains $\lambda^{(\nu)}$ which might seem to involve some difficulty, since $x^{(\nu)}$ cannot be found until we know $\lambda^{(\nu)}$. However, if we take the inner product of both sides of Eq. (4.54) with $x^{(0)}$, we get

$$(x^{(0)}, [A_0 - \lambda^{(0)}] x^{(\nu)}) = \sum_{\mu=1}^{\nu} \lambda^{(\mu)} (x^{(0)}, x^{(\nu-\mu)}) - (x^{(0)}, A_1 x^{(\nu-1)}) .$$

Using Eq. (4.50) and the fact that A_0 is Hermitian, we see that the left-hand side vanishes, so

$$\lambda^{(\nu)} = (x^{(0)}, A_1 x^{(\nu-1)}) - \sum_{\mu=1}^{\nu-1} \lambda^{(\mu)} (x^{(0)}, x^{(\nu-\mu)})$$

for $\nu = 1, 2, \cdots$. Hence $\lambda^{(\nu)}$ is determined in terms of the $\lambda^{(i)}$ and $x^{(i)}$ of orders *lower* than ν.

Thus, in principle, we should be able to solve the hierarchy of equations implied by Eq. (4.54), with the first $(n - 1)$ equations providing the input for

the nth equation. The first few equations of the sequence are

$$[A_0 - \lambda^{(0)}]x^{(0)} = 0 ,$$

$$[A_0 - \lambda^{(0)}]x^{(1)} = -[A_1 - \lambda^{(1)}]x^{(0)} , \tag{4.55a}$$

$$[A_0 - \lambda^{(0)}]x^{(2)} = -[A_1 - \lambda^{(1)}]x^{(1)} + \lambda^{(2)}x^{(0)} , \tag{4.55b}$$

$$[A_0 - \lambda^{(0)}]x^{(3)} = -[A_1 - \lambda^{(1)}]x^{(2)} + \lambda^{(2)}x^{(1)} + \lambda^{(3)}x^{(0)} . \tag{4.55c}$$

The first equation is just Eq. (4.50), and provides the starting point for solving all the others, order by order.

All of the Eqs. (4.55) except for the first are of the type $Hx = h$, discussed in Theorem 3.17. Since $\lambda^{(0)}$ is an eigenvalue of A_0, $\det H = 0$. Therefore h must lie entirely within the subspace spanned by the eigenvectors of H belonging to nonzero eigenvalues, that is, h must be orthogonal to all the eigenvectors of H belonging to a zero eigenvalue. In this case, h must be orthogonal to all eigenvectors of A_0 belonging to $\lambda^{(0)}$.

It is here that we see the first practical difference between the case of degenerate eigenvalues and nondegenerate ones. If we have only nondegenerate eigenvalues in mind, then there is only *one* eigenvector to which h must be orthogonal. We shall discuss this case in some detail first, since its bookkeeping is simplest.

Let us assume, then, that the spectrum of A_0 is nondegenerate, and imagine that we have determined a particular $x^{(0)}$ from Eq. (4.50). The discussion of the preceding paragraph tells us that Eq. (4.55a) can give a solution for $x^{(1)}$ only if

$$(x^{(0)}, [A_1 - \lambda^{(1)}]x^{(0)}) = 0 .$$

Since we may choose $x^{(0)}$ to be normalized to 1, this yields

$$\lambda^{(1)} = (x^{(0)}, A_1x^{(0)}) . \tag{4.56a}$$

If we apply the same requirement to Eq. (4.55b), we have

$$\lambda^{(2)} = (x^{(0)}, [A_1 - \lambda^{(1)}]x^{(1)}) . \tag{4.56b}$$

This gives us $\lambda^{(2)}$ in terms of quantities which are, in principle, known. Similarly, Eq. (4.55c) has a well-defined solution provided that

$$\lambda^{(3)} = (x^{(0)}, [A_1 - \lambda^{(1)}]x^{(2)}) - \lambda^{(2)}(x^{(0)}, x^{(1)}) .$$

This is just like our equations for $\lambda^{(1)}$ and $\lambda^{(2)}$, but here it turns out that we can do better. Using the Hermiticity of A_1, we can write

$$(x^{(0)}, [A_1 - \lambda^{(1)}]x^{(2)}) = ([A_1 - \lambda^{(1)}]x^{(0)}, x^{(2)}) .$$

Using Eq. (4.55a), we get

$$\begin{aligned}(x^{(0)}, [A_1 - \lambda^{(1)}]x^{(2)}) &= -([A_0 - \lambda^{(0)}]x^{(1)}, x^{(2)}) \\ &= -(x^{(1)}, [A_0 - \lambda^{(0)}]x^{(2)}) ,\end{aligned}$$

again using the Hermiticity of A_0. Finally, Eq. (4.55b) yields

$$(x^{(0)}, [A_1 - \lambda^{(1)}]x^{(2)}) = (x^{(1)}, [A_1 - \lambda^{(1)}]x^{(1)}) - \lambda^{(2)}(x^{(1)}, x^{(0)}) .$$

Putting this into the equation for $\lambda^{(3)}$, we obtain

$$\lambda^{(3)} = (x^{(1)}, [A_1 - \lambda^{(1)}]x^{(1)}) - \lambda^{(2)} [(x^{(0)}, x^{(1)}) + (x^{(1)}, x^{(0)})] . \tag{4.56c}$$

Thus $\lambda^{(3)}$ is determined only by $x^{(0)}$ and $x^{(1)}$, but not by $x^{(2)}$. This is a general feature of perturbation theory. In fact, each $x^{(i)}$ determines two terms in the expansion of λ. We have already seen that $\lambda^{(0)}$ and $\lambda^{(1)}$ are found by just knowing $x^{(0)}$, and $\lambda^{(2)}$ and $\lambda^{(3)}$ are determined by $x^{(0)}$ and $x^{(1)}$. It is a fairly simple exercise to show that $\lambda^{(4)}$ and $\lambda^{(5)}$ can be obtained from $x^{(0)}$, $x^{(1)}$, and $x^{(2)}$, and similarly for terms of higher order.

We may simplify Eqs. (4.56) slightly by noticing that we may choose $(x^{(0)}, x^{(i)}) = 0$, for $i > 0$. This is true because if any one of Eqs. (4.55) for $x^{(i)}$ has some solution $y^{(i)}$, then $y^{(i)} + ax^{(0)}$ is also a solution, because $(A_0 - \lambda^{(0)})x^{(0)} = 0$. Therefore we can always arrange that

$$(x^{(0)}, x^{(i)}) = (x^{(0)}, y^{(i)} + ax^{(0)}) = (x^{(0)}, y^{(i)}) + a = 0$$

by taking as the solutions the vectors $y^{(i)} + ax^{(0)}$, where $a = -(x^0, y^{(i)})$. Then Eqs. (4.56) become

$$\lambda^{(1)} = (x^{(0)}, A_1x^{(0)}) , \tag{4.57a}$$

$$\lambda^{(2)} = (x^{(0)}, A_1x^{(1)}) , \tag{4.57b}$$

$$\lambda^{(3)} = (x^{(1)}, [A_1 - \lambda^{(1)}]x^{(1)}) . \tag{4.57c}$$

We should emphasize that these values of $\lambda^{(1)}$, $\lambda^{(2)}$, and $\lambda^{(3)}$ are obtained by requiring that well-defined solutions exist for Eqs. (4.55). With the λ's now given by Eqs. (4.57), we are guaranteed that such solutions do indeed exist. The only task remaining is to see about solving Eqs. (4.55), for which there are several methods available.

The most obvious method would be to solve the matrix equations directly. However, since the main application of perturbation theory is to operators on *infinite*-dimensional spaces, where there is no simple matrix representation, we shall say no more about this direct method other than that it exists and is in principle easy to use on a finite-dimensional space.

If we were fortunate enough to know the complete set of eigenfunctions and eigenvalues of the Hermitian operator A_0, there would be another possibility: we could expand the $x_n^{(i)}$ $(i = 1, 2, \cdots)$ in terms of the eigenfunctions of A_0, say $x_m^{(0)}$. We have now restored the subscript n which indexes *different* eigenvalues and eigenvectors. As an example, we compute $x_n^{(1)}$ in terms of the $x_m^{(0)}$; it is a simple matter to extend this technique to obtain $x_n^{(i)}$, for $i \geq 2$.

First, we assume that $m \neq n$, and take the inner product of both sides of Eq. (4.55a) on the left with $x_m^{(0)}$. Thus we get

$$(x_m^0, [A_0 - \lambda_n^{(0)}]x_n^{(1)}) = -(x_m^{(0)}, [A_1 - \lambda_n^{(1)}]x_n^{(0)}) . \tag{4.58}$$

Since the eigenfunctions of A_0 form an orthonormal set, the right-hand side of Eq. (4.58) is just

$$-(x_m^{(0)}, A_1 x_n^{(0)}) .$$

On the left-hand side of Eq. (4.58), we use the Hermiticity of A_0 and the fact that $x_m^{(0)}$ is an eigenfunction of A_0 with eigenvalue $\lambda_m^{(0)}$ to write it as

$$(\lambda_m^{(0)} - \lambda_n^{(0)})(x_m^{(0)}, x_n^{(1)}) .$$

Then Eq. (4.58) becomes

$$(\lambda_n^{(0)} - \lambda_m^{(0)})(x_m^{(0)}, x_n^{(1)}) = (x_m^{(0)}, A_1 x_n^{(0)}) , \qquad m \neq n .$$

Since there is no degeneracy, $\lambda_n^{(0)} \neq \lambda_m^{(0)}$ so long as $n \neq m$. Therefore

$$(x_m^{(0)}, x_n^{(1)}) = \frac{(x_m^{(0)}, A_1 x_n^{(0)})}{\lambda_n^{(0)} - \lambda_m^{(0)}} \tag{4.59}$$

for $m \neq n$. For $m = n$, we have already decided to take $(x_n^{(0)}, x_n^{(1)}) = 0$. We now use the completeness of the eigenfunctions of A_0 to write

$$x_n^{(1)} = \sum_m (x_m^{(0)}, x_n^{(1)}) x_m^{(0)} .$$

For our particular case, this is just

$$x_n^{(1)} = \sum_{m \neq n} \frac{(x_m^{(0)}, A_1 x_n^{(0)})}{\lambda_n^{(0)} - \lambda_m^{(0)}} x_m^{(0)} . \tag{4.60}$$

This gives us $x_n^{(1)}$; the same method applied to Eqs. (4.55b) and (4.55c) would yield $x_n^{(2)}$ and $x_n^{(3)}$. Note that according to Eqs. (4.57b) and (4.57c), if we know $x_n^{(0)}$ and $x_n^{(1)}$, we can find $\lambda_n^{(2)}$ and $\lambda_n^{(3)}$.

We can illustrate all these operations with the following simple example. We showed in Example 1 of Section 3.10 that the Hermitian Pauli matrix

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

has eigenvalues $\lambda_1^{(0)} = +1$, $\lambda_2^{(0)} = -1$ and normalized eigenvectors

$$x_1^{(0)} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad x_2^{(0)} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} .$$

Using nondegenerate perturbation theory, we can compute the eigenvectors of the matrix

$$\sigma = \begin{bmatrix} \epsilon & 1-\epsilon \\ 1-\epsilon & 3\epsilon \end{bmatrix} = \sigma_x + \epsilon \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}$$

through first order and thus obtain the eigenvalues through third order in ϵ.

Since

$$A_1 = \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix},$$

$\lambda^{(1)}$ is given by

$$\lambda_1^{(1)} = (x_1^{(0)}, A_1 x_1^{(0)}) = \frac{1}{\sqrt{2}}[1 \quad 1]\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\frac{1}{\sqrt{2}}$$
$$= \frac{1}{2}[1 \quad 1]\begin{bmatrix} 0 \\ 2 \end{bmatrix} = 1\,,$$

$$\lambda_2^{(1)} = (x_2^{(0)}, A_1 x_2^{(0)}) = \frac{1}{\sqrt{2}}[1 \quad -1]\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}\frac{1}{\sqrt{2}}$$
$$= \frac{1}{2}[1 \quad -1]\begin{bmatrix} 2 \\ -4 \end{bmatrix} = 3\,.$$

Therefore, through first order,

$$\lambda_1 = 1 + \epsilon\,, \qquad \lambda_2 = -1 + 3\epsilon\,.$$

Now, to get the correction to the eigenvector through first order, we use Eq. (4.60):

$$x_1^{(1)} = \sum_{m\neq 1} \frac{(x_m^{(0)}, A_1 x_1^{(0)})}{\lambda_1^{(0)} - \lambda_m^{(0)}} x_m^{(0)} = \frac{(x_2^{(0)}, A_1 x_1^{(0)})}{\lambda_1^{(0)} - \lambda_2^{(0)}} x_2^{(0)}$$
$$= \frac{1}{1-(-1)}\frac{1}{\sqrt{2}}[1 \quad -1]\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\frac{1}{\sqrt{2}} x_2^{(0)}$$
$$= -\frac{1}{2}x_2^{(0)} = -\frac{1}{2\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Similarly,

$$x_2^{(1)} = +\frac{1}{2}x_1^{(0)} = \frac{1}{2\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Then, through first order in ϵ,

$$x_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 - \epsilon/2 \\ 1 + \epsilon/2 \end{bmatrix}, \qquad x_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 + \epsilon/2 \\ -1 + \epsilon/2 \end{bmatrix}. \tag{4.61}$$

With $x_1^{(1)}$ and $x_2^{(1)}$, we can now find the second and third order corrections to λ_1 and λ_2. Using Eq. (4.57b), we get

$$\lambda_1^{(2)} = (x_1^{(0)}, A_1 x_1^{(1)}) = \frac{1}{\sqrt{2}}[1 \quad 1]\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}\frac{(-1)}{2\sqrt{2}}$$
$$= -\frac{1}{4}[1 \quad 1]\begin{bmatrix} 2 \\ -4 \end{bmatrix} = +\frac{1}{2}\,,$$

$$\lambda_2^{(2)} = (x_2^{(0)}, A_1 x_2^{(1)}) = \frac{1}{\sqrt{2}}[1 \quad -1]\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\frac{1}{2\sqrt{2}}$$
$$= \frac{1}{4}[1 \quad -1]\begin{bmatrix} 0 \\ 2 \end{bmatrix} = -\frac{1}{2}\,.$$

Also, using Eqs. (4.57c), we obtain

$$\begin{aligned}\lambda_1^{(3)} &= (x_1^{(1)}, A_1 x_1^{(1)}) - \lambda_1^{(1)}(x_1^{(1)}, x_1^{(1)}) \\ &= \frac{-1}{2\sqrt{2}}[1 \quad -1]\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}\frac{(-1)}{2\sqrt{2}} \\ &\quad - 1\,\frac{(-1)}{2\sqrt{2}}(1 \quad -1)\begin{bmatrix} 1 \\ -1 \end{bmatrix}\frac{(-1)}{2\sqrt{2}} \\ &= \frac{3}{4} - \frac{1}{4} = \frac{1}{2}\,.\end{aligned}$$

Similarly, $\lambda_2^{(3)} = -\frac{1}{2}$. Combining all these results, we find, through third order in ϵ, that

$$\begin{aligned}\lambda_1 &= 1 + \epsilon + \tfrac{1}{2}\epsilon^2 + \tfrac{1}{2}\epsilon^3 + \cdots, \\ \lambda_2 &= -1 + 3\epsilon - \tfrac{1}{2}\epsilon^2 - \tfrac{1}{2}\epsilon^3 + \cdots\end{aligned} \tag{4.62}$$

Our final results, Eqs. (4.61) and (4.62), can easily be checked by diagonalizing A directly, that is, by solving

$$\begin{bmatrix} \epsilon & 1-\epsilon \\ 1-\epsilon & 3\epsilon \end{bmatrix} x_n = \lambda_n x_n\,.$$

Solving the characteristic equation gives

$$\lambda_1 = 2\epsilon + \sqrt{1 - 2(\epsilon - \epsilon^2)}\,, \qquad \lambda_2 = 2\epsilon - \sqrt{1 - 2(\epsilon - \epsilon^2)}\,,$$

and if we expand these in ϵ by using the binomial theorem, we reproduce Eq. (4.62) exactly through third order. We leave it as an exercise to show that the eigenvectors of A, which can now be calculated exactly, since λ_1 and λ_2 are known exactly, agree through first order with Eq. (4.61).

We see that in this case the radius of convergence of perturbation theory is given by the radius of convergence of the binomial expansion of $\sqrt{1 - 2(\epsilon - \epsilon^2)}$. In general, it is very difficult to give a rigorous discussion of this type of perturbation theory, which is often called Rayleigh-Schrodinger perturbation theory, after its principal developers.

4.12 PERTURBATION THEORY—DEGENERATE CASE

In the last section we found that if the zero-order operator, A_0, has a nondegenerate spectrum, the basic results of perturbation theory follow straightforwardly from the completeness of the eigenfunctions of A_0 and Theorem 3.17. We shall now show that the same situation holds when A_0 possesses degenerate eigenvalues. The only real difficulty in this case will be a notational one.

It is tempting to think that we should immediately replace Eq. (4.50) by

$$[A_0 - \lambda_n^{(0)}]\, x_{n,i}^{(0)} = 0\,,$$

where $i = 1, 2, \cdots, \mu_n$, μ_n being the multiplicity of the nth eigenvalue, and then proceed to rewrite Eq. (4.55a) as

$$[A_0 - \lambda_n^{(0)}]\, x_{n,i}^{(1)} = -[A_1 - \lambda_{n,i}^{(1)}]\, x_{n,i}^{(0)} .$$

But at this point we run into the difficulty that, for a solution to exist, Theorem 3.17 requires not only that

$$(x_{n,i}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]\, x_{n,i}^{(0)}) = 0 ,$$

but also, since $\lambda_n^{(0)}$ has multiplicity μ_n, that

$$(x_{n,j}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]\, x_{n,i}^{(0)}) = 0 \qquad \text{for all } j .$$

If $j \neq i$, this last equation becomes

$$(x_{n,i}^{(0)}, A_1 x_{n,j}^{(0)}) = 0 ,$$

which looks like a set of restrictions on our perturbation operator, A_1. This is clearly an inadmissible situation. At this point, however, the degeneracy difficulty actually comes to our rescue. What we have forgotten is that the statement "$x_{n,i}^{(0)}$ $(i = 1, 2, \cdots, \mu_n)$ are *the* eigenvectors of A_0 corresponding to $\lambda_n^{(0)}$" is not quite precise, because, in fact, *any* linear combination of the $x_{n,i}^{(0)}$ $(i = 1, 2, \cdots \mu_n)$ is also a perfectly good zero-order solution.

Therefore let us take as zero-order solutions the set $y_{n,i}^{(0)}$ $(i = 1, 2, \cdots, \mu_n)$, where

$$y_{n,i}^{(0)} = \sum_{j=1}^{\mu_n} a_j^{(i)} x_{n,j}^{(0)} . \tag{4.63}$$

We shall see that the $a_i^{(j)}$ can be determined in such a way that we shall not run into difficulty in calculating corrections to $\lambda_n^{(0)}$.

If we require that $(y_{n,j}^{(0)}, y_{n,k}^{(0)}) = \delta_{jk}$, then we must have

$$\sum_i a_i^{(j)*} a_i^{(k)} = \delta_{jk} . \tag{4.64}$$

Thus, instead of Eq. (4.50), we write

$$[A_0 - \lambda_n^{(0)}] y_{n,i}^{(0)} = 0 \tag{4.50'}$$

and replace Eq. (4.52) by

$$\begin{aligned} \lambda_{n,i} &= \lambda_n^{(0)} + \epsilon\lambda_{n,i}^{(1)} + \epsilon^2\lambda_{n,i}^{(2)} + \cdots, \\ y_{n,i} &= y_{n,i}^{(0)} + \epsilon y_{n,i}^{(1)} + \epsilon^2 y_{n,i}^{(2)} + \cdots, \end{aligned} \tag{4.52'}$$

where $y_{n,i}^{(0)}$ is given by Eq. (4.63). What we are saying is that in solving Eq. (4.50), we may well have found a set of eigenfunctions $x_{n,i}^{(0)}$ which are acceptable as solutions of Eq. (4.50), but, are nevertheless not ideal for handling higher order corrections. We want to find those linear combinations of the $x_{n,i}^{(0)}$ which will do the job. Also, in labeling our eigenvalues, we have included a degeneracy index i on all terms except $\lambda_n^{(0)}$, since we expect that an arbitrary

perturbation will split the degeneracy to some extent. Using Eq. (4.52′), we find, just as in the last section, the analogs of Eqs. (4.55):

$$[A_0 - \lambda_n^{(0)}]y_{n,i}^{(1)} = -[A_1 - \lambda_{n,i}^{(1)}]y_{n,i}^{(0)}\,, \tag{4.55a′}$$

$$[A_0 - \lambda_n^{(0)}]y_{n,i}^{(2)} = -[A_1 - \lambda_{n,i}^{(1)}]y_{n,i}^{(1)} + \lambda_{n,i}^{(2)}y_{n,i}^{(0)}\,, \tag{4.55b′}$$

$$[A_0 - \lambda_n^{(0)}]y_{n,i}^{(3)} = -[A_1 - \lambda_{n,i}^{(1)}]y_{n,i}^{(2)} + \lambda_{n,i}^{(2)}y_{n,i}^{(1)} + \lambda_{n,i}^{(3)}y_{n,i}^{(0)}\,. \tag{4.55c′}$$

We now reap the rewards of our reformulation of the problem. For a solution to Eq. (4.55a′) to exist, we require that the vector on the right-hand side of Eq. (4.55a′) have no component in the subspace spanned by the set $\{x_{n,i}^{(0)}\}$ (which is, of course, the same as the subspace spanned by the $\{y_{n,i}^{(0)}\}$). In other words, we require that

$$(x_{n,j}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]y_{n,i}^{(0)}) = 0 \qquad \text{for all } i, j.$$

But by Eq. (4.63), this is just

$$\sum_k (x_{n,j}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]x_{n,k}^{(0)})a_k^{(i)} = 0$$

or

$$\sum_k [(x_{n,j}^{(0)}, A_1x_{n,k}^{(0)}) - \lambda_{n,i}^{(1)}\delta_{jk}]a_k^{(i)} = 0\,, \tag{4.65a}$$

which is a simple eigenvalue problem of the type we have been discussing in this chapter. This is particularly clear if we define a matrix M whose jk element is given by

$$M_{jk} = (x_{n,j}^{(0)}, A_1x_{n,k}^{(0)})$$

and a vector $a^{(i)}$ whose kth element is $a_k^{(i)}$. Then Eq. (4.65a) can be written as

$$(M - \lambda_{n,i}^{(1)})a^{(i)} = 0\,. \tag{4.65b}$$

Since A_1 is Hermitian, so is M, and Eq. (4.65b) is therefore a Hermitian eigenvalue problem. Thus $\lambda_{n,i}^{(i)}$ will be real, and the $a^{(i)}$ can be chosen so that

$$(a^{(i)}, a^{(j)}) = \delta_{ij}\,,$$

in keeping with the requirement of Eq. (4.64). Thus, all at once, we have: (1) found the first corrections to $\lambda_n^{(0)}$, (2) determined the correct linear combinations of the $x_{n,i}^{(0)}$, and (3) guaranteed that Eq. (4.55a′) will have a solution.

In summary, our procedure is the following: We calculate the matrix of A_1 in the subspace spanned by the set $\{x_{n,i}^{(0)}, i = 1, 2, \cdots, \mu_n\}$. We then find the eigenvalues and orthonormal eigenvectors of this $\mu_n \times \mu_n$ matrix. The eigenvalues are the first-order corrections to $\lambda_n^{(0)}$, and the eigenvectors provide the correct linear combinations of the $y_{n,i}^{(0)}$ from Eq. (4.63).

At this point Eq. (4.55a′) can be solved for $y_{n,i}^{(1)}$. For example, if we know *all* the eigenvectors and eigenvalues of A_0, then by the expansion method discussed in Section 4.11, we get, for $m \neq n$,

$$(x_{m,i}^{(0)}, y_{n,j}^{(1)}) = \frac{(x_{m,i}^{(0)}, A_1y_{n,j}^{(0)})}{\lambda_n^{(0)} - \lambda_m^{(0)}}\,. \tag{4.66}$$

Note, however, that we do not yet have the μ_n components of $y_{n,j}^{(1)}$ along $x_{n,i}^{(0)}$ ($i = 1, 2, \cdots, \mu_n$), that is, we do not know $(x_{n,i}^{(0)}, y_{n,j}^{(1)})$. Therefore we cannot yet use the expansion theorem.

In the nondegenerate case, where only one component of the first-order eigenvector was missing, we found that we could, with no risk, set the missing component equal to zero. This is reasonable since we have one loose degree of freedom associated with the normalization of the exact eigenvector (which in physical problems is usually chosen to be normalized to unity). But in the present degenerate case, we will have μ_n components missing, and we clearly cannot get away with setting them all equal to zero.

However, it turns out that the same situation which enabled us to resolve the ambiguity in the choice of zero-order wavefunction will rescue us here. When we go to Eq. (4.55b′) to determine $\lambda_{n,i}^{(2)}$, we shall find in the process the missing components of $y_{n,i}^{(1)}$.

For Eq. (4.55b′) to have a solution, the vectors on the right-hand side must have no component lying in the subspace spanned by the $\{x_{n,j}^{(0)}, j = 1, 2, \cdots, \mu_n\}$, that is,

$$(x_{n,j}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]y_{n,i}^{(1)}) - \lambda_{n,i}^{(2)}(x_{n,j}^{(0)}, y_{n,i}^{(0)}) = 0$$

for all i and j. Using Eq. (4.63), we obtain

$$a_j^{(i)}\lambda_{n,i}^{(2)} = (x_{n,j}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]y_{n,i}^{(1)}) \,. \tag{4.67}$$

Now we split $y_{n,i}^{(1)}$ into two parts,

$$y_{n,i}^{(1)} = f_{n,i} + g_{n,i} \,,$$

when $f_{n,i}$ is the unknown part of $y_{n,i}^{(1)}$ lying in the subspace spanned by the set $\{x_{n,j}^{(0)}, j = 1, 2, \cdots \mu_n\}$ and $g_{n,i}$ is the rest, which according to Eq. (4.66) we already know. In this way, Eq. (4.67) can be rewritten as

$$(x_{n,j}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]f_{n,i}) = -(x_{n,j}^{(0)}, A_1 g_{n,i}) + a_j^{(i)}\lambda_{n,i}^{(2)} \,. \tag{4.68}$$

Finally, we write $f_{n,i}$ as

$$f_{n,i} = \sum_k b_k^{(i)} x_{n,k}^{(0)} \,, \tag{4.69}$$

and substituting into Eq. (4.68), we get

$$\sum_k (x_{n,j}^{(0)}, [A_1 - \lambda_{n,i}^{(1)}]\, x_{n,k}^{(0)}) b_k^{(i)} = \lambda_{n,i}^{(2)} a_j^{(i)} - (x_{n,j}^{(0)}, A_1 g_{n,i}) \qquad \text{for all } i, j. \tag{4.70a}$$

If the reader will look carefully at Eq. (4.70a), he will see, through the forest of subscripts and superscripts, that it is a simple matrix equation. In fact, if we define a vector $b^{(i)}$ whose components are the μ_n scalars $b_k^{(i)}$ and a vector $a^{(i)}$ whose components are the $a_k^{(i)}$, then Eq. (4.70a) can be written as

$$(M - \lambda_{n,i}^{(1)})b^{(i)} = \lambda_{n,i}^{(2)} a^{(i)} - c^{(i)} \tag{4.70b}$$

for all i, where M is the matrix, defined earlier, whose jk element is $(x_{n,j}^{(0)}, A_1 x_{n,k}^{(0)})$ and $c^{(i)}$ is a vector whose jth component is $(x_{n,j}^{(0)}, A_1 g_{n,i})$. Thus the vectors $b^{(i)}$ are determined by solving a simple matrix equation of the type $Hx = h$. Again

according to Eq. (4.70b), H is singular, since it was precisely by diagonalizing the matrix on the left-hand side of Eq. (4.70b) that we determined the $\lambda_{n,i}^{(1)}$ in Eq. (4.65b). Hence our very useful Theorem 3.17 tells us that for a solution to exist, the right-hand side of Eq. (4.70b) must be orthogonal to all the eigenvectors of Eq. (4.65b) belonging to eigenvalue $\lambda_{n,i}^{(1)}$. Let us suppose that perturbation completely removed the degeneracy in first order so none of the $\lambda_{n,i}^{(1)}$ ($i = 1, 2, \cdots, \mu_n$) are equal. Then the only eigenvector belonging to $\lambda_{n,i}^{(1)}$ is $a^{(i)}$; by the above remarks, a solution to Eq. (4-70b) exists if

$$\lambda_{n,i}^{(2)}(a^{(i)}, a^{(i)}) - (a^{(i)}, c^{(i)}) = 0$$

or in component form,

$$\sum_j a_j^{(i)*}[a_j^{(i)}\lambda_{n,i}^{(2)} - (x_{n,j}^{(0)}, A_1 g_{n,i})] = 0 .$$

Making use of Eq. (4.63) and Eq. (4.64), this reduces to

$$\lambda_{n,i}^{(2)} = (y_{n,i}^{(0)}, A_1 g_{n,i}) . \tag{4.71}$$

If we know all the eigenvectors of A_0 and the corresponding eigenvalues, then by Eq. (4.66),

$$g_{n,j} = \sum_{m \neq n} \sum_{i=1}^{\mu_m} \frac{(x_{m,i}^{(0)}, A_1 y_{n,j}^{(0)})}{\lambda_n^{(0)} - \lambda_m^{(0)}} x_{m,i}^{(0)} , \tag{4.66$'$}$$

so $\lambda_{n,i}^{(2)}$ is determined. We should emphasize that Eq. (4.71) bears a very close resemblance to Eq. (4.57b) of the last section.

Here again, the equation which determines $\lambda_{n,i}^{(2)}$ guarantees the existence of solutions to Eq. (4.70b) for $b^{(i)}$ ($i = 1, 2, \cdots, \mu_n$), so the missing components of $y_{n,i}^{(1)}$ are found, although the component of $b^{(i)}$ in the direction $a^{(i)}$ is, by Theorem 3.17, undetermined. (It is simple to show that this corresponds to the fact that $(y_{n,i}^{(0)}, y_{n,i}^{(1)})$ is indeterminate.) It is often convenient to pick this inner product equal to zero, which is equivalent to taking $b^{(i)}$ orthogonal to $a^{(i)}$. Finally, with $\lambda_{n,i}^{(2)}$ and $b^{(i)}$ determined in this manner for all i, Theorem 3.17 guarantees that a solution exists to Eq. (4.55b$'$), so $y_{n,i}^{(2)}$ is now known, again with indeterminacy along the directions defined by $\{x_{n,i}^{(0)}, i = 1, 2, \cdots, \mu_n\}$. It is possible, but tedious, to carry on to Eq. (4.51c$'$), remove the indeterminacy in $y_{n,i}^{(2)}$, and at the same time find $\lambda_{n,i}^{(3)}$.

In going through second order, we have assumed that the degeneracy was completely removed in first order. This assumption is *unnecessary*, and we leave it as an exercise in notation for the interested reader to show that if there is some degeneracy left after first order, we must solve an eigenvalue problem for those $\lambda^{(2)}$'s which correspond to degenerate $\lambda^{(1)}$'s, instead of a simple equation for $\lambda_{n,i}^{(2)}$. Again we emphasize that this is a purely notational problem, identical to that encountered in going from nondegenerate to degenerate perturbation theory. In principle, the entire Rayleigh-Schrodinger perturbation theory stems from the existence of an expansion of the type given in Eq. (4.52), followed by repeated application of Theorem 3.17.

Example. As an application of degenerate perturbation theory, consider the linear operator

$$A_0 = \frac{p_x^2}{2m} + \frac{1}{2} m\omega_0^2 x^2 + \frac{p_y^2}{2m} + \frac{1}{2} m\omega_0^2 y^2 . \tag{4.72}$$

This is a sum of two independent linear operators of the type discussed in Section 3.10. We impose the commutation relations

$$[x, p_x] = i\hbar I , \qquad [y, p_y] = i\hbar I ,$$

and all other commutators (e.g., $[x, p_y]$) vanish. As in Section 3.10, we define

$$\begin{aligned} a_x &= (xm\omega_0 + ip_x)(2m\omega_0\hbar)^{-1/2} , & a_y &= (ym\omega_0 + ip_y)(2m\omega_0\hbar)^{-1/2} , \\ a_x^\dagger &= (xm\omega_0 - ip_x)(2m\omega_0\hbar)^{-1/2} , & a_y^\dagger &= (ym\omega_0 - ip_y)(2m\omega_0\hbar)^{-1/2} . \end{aligned} \tag{4.73}$$

An easy computation, like that leading to Eq. (3.47) shows that

$$A_0 = \hbar\omega_0(a_x^\dagger a_x + a_y^\dagger a_y + I) \tag{4.74}$$

and

$$[a_x, a_x^\dagger] = I = [a_y, a_y^\dagger] ,$$
$$[a_x, a_y] = [a_x^\dagger, a_y^\dagger] = [a_x, a_y^\dagger] = [a_x^\dagger, a_y] = 0 .$$

It follows that

$$[a_i^\dagger, A_0] = -\hbar\omega_0 a_i^\dagger , \qquad [a_i, A_0] = \hbar\omega_0 a_i ,$$

for $i = x$ and y. By methods completely identical to those of Section 3.10, it then follows that the eigenfunctions of A_0 are given by

$$\phi_{n,m} = (a_x^\dagger)^n (a_y^\dagger)^m \phi_{0,0}/\sqrt{n!m!} , \tag{4.75a}$$

with corresponding eigenvalues

$$\lambda_N^{(0)} = \hbar\omega_0(n + m + 1) = \hbar\omega_0(N + 1) . \tag{4.75b}$$

If we require that $(\phi_{0,0}, \phi_{0,0}) = 1$, then

$$(\phi_{n,m}, \phi_{\nu,\mu}) = \delta_{n\nu}\delta_{m\mu} .$$

Also, with these definitions, we have

$$\begin{aligned} a_x^\dagger\phi_{n,m} &= (n + 1)^{1/2}\phi_{n+1,m} , & a_y^\dagger\phi_{n,m} &= (m + 1)^{1/2}\phi_{n,m+1} , \\ a_x\phi_{n,m} &= (n)^{1/2}\phi_{n-1,m} , & a_y\phi_{n,m} &= (m)^{1/2}\phi_{n,m-1} . \end{aligned} \tag{4.76}$$

Now suppose that A_0 is perturbed by a term of the form

$$\epsilon A_1 = \epsilon m\omega_0^2 xy .$$

In terms of the creation and annihilation operators, this is

$$\epsilon A_1 = \tfrac{1}{2}\epsilon\hbar\omega_0(a_x^\dagger a_y + a_x a_y^\dagger + a_x^\dagger a_y^\dagger + a_x a_y) . \tag{4.77}$$

The lowest eigenvalue of A_0 corresponds to $n = m = 0$: $\lambda_{0,0} = \hbar\omega_0$; it is nondegenerate. Thus the theory of Section 4.11 is applicable, and we leave it for

the reader to show that calculating all terms through ϵ^2, we get

$$\begin{aligned} \lambda_{0,0} &= \lambda_{0,0}^{(0)} + \epsilon\lambda_{0,0}^{(1)} + \epsilon^2\lambda_{0,0}^{(2)} + \cdots \\ \lambda_{0,0} &= \hbar\omega_0(1 - \epsilon^2/8 + \cdots) . \end{aligned} \tag{4.78}$$

The next eigenvalue is doubly degenerate, since according to Eqs. (4.75), there are two eigenvectors corresponding to the eigenvalue $2\hbar\omega_0$, namely,

$$\phi_{1,0} = a_x^\dagger\phi_{0,0} , \qquad \phi_{0,1} = a_y^\dagger\phi_{0,0} .$$

Therefore, to obtain corrections to this eigenvalue, we need degenerate perturbation theory. In keeping with the notation established earlier in this section, we write

$$x_{1,1}^{(0)} = \phi_{1,0} , \qquad x_{1,2}^{(0)} = \phi_{0,1}^{(0)} , \tag{4.79}$$

According to Eq. (4.65), we must find the matrix of A_1 in the subspace spanned by $x_{1,1}^{(0)}$ and $x_{1,2}^{(0)}$. Using Eqs. (4.76), (4.77), and (4.79), we find for this matrix

$$\tfrac{1}{2}\hbar\omega_0\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} .$$

Following Eq. (4.65b), we diagonalize it to get $\lambda_{1,1}^{(1)} = \frac{1}{2}\hbar\omega_0$, $\lambda_{1,2}^{(1)} = -\frac{1}{2}\hbar\omega_0$, with corresponding eigenvectors,

$$a^{(1)} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} , \qquad a^{(2)} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix} .$$

Equation (4.63) then gives us the correct zero-order solutions:

$$\begin{aligned} y_{1,1}^{(0)} &= \frac{1}{\sqrt{2}}(x_{1,1}^{(0)} + x_{1,2}^{(0)}) = \frac{1}{\sqrt{2}}(\phi_{1,0} + \phi_{0,1}) , \\ y_{1,2}^{(0)} &= \frac{1}{\sqrt{2}}(x_{1,1}^{(0)} - x_{1,2}^{(0)}) = \frac{1}{\sqrt{2}}(\phi_{1,0} - \phi_{0,1}) . \end{aligned}$$

These give us through first order in ϵ:

$$\lambda_{1,1} = 2\hbar\omega_0 + \tfrac{1}{2}\epsilon\hbar\omega_0 , \qquad \lambda_{1,2} = 2\hbar\omega_0 - \tfrac{1}{2}\epsilon\hbar\omega_0 .$$

With these results established, we now can solve Eq. (4.55a′) for $y_{1,1}^{(1)}$ and $y_{1,2}^{(1)}$. Since we know all the eigenvectors and eigenvalues of A_0, we use Eq. (4.66) to determine $g_{1,1}$ and $g_{1,2}$, the components of $y_{1,1}^{(1)}$ and $y_{1,2}^{(1)}$ which do not lie in the subspace spanned by $y_{1,1}^{(0)}$ and $y_{1,2}^{(0)}$. According to Eq. (4.71), the g's will determine $\lambda_{1,1}^{(2)}$ and $\lambda_{1,2}^{(2)}$. Using Eqs. (4.66′), (4.75), (4.76), (4.77), and (4.79), we get

$$g_{1,1} = \tfrac{1}{4}(\phi_{2,1} + \phi_{1,2}) , \qquad g_{1,2} = \tfrac{1}{4}(\phi_{2,1} - \phi_{1,2}) .$$

Thus, according to Eq. (4.71),

$$\begin{aligned} \lambda_{1,1}^{(2)} &= (y_{1,1}^{(0)}, A_1 g_{1,1}) \\ &= \frac{1}{8\sqrt{2}}\hbar\omega_0([\phi_{1,0} + \phi_{0,1}], [a_x^\dagger a_y + a_x a_y^\dagger - a_x^\dagger a_y^\dagger - a_x a_y][\phi_{2,1} + \phi_{1,2}]) \\ &= -\tfrac{1}{4}\hbar\omega_0 = \lambda_{1,2}^{(2)} . \end{aligned}$$

Through second order in ϵ, we have

$$\begin{aligned} \lambda_{1,1} &= \hbar\omega_0(2 + \tfrac{1}{2}\epsilon - \tfrac{1}{4}\epsilon^2 + \cdots) , \\ \lambda_{1,2} &= \hbar\omega_0(2 - \tfrac{1}{2}\epsilon - \tfrac{1}{4}\epsilon^2 + \cdots) , \end{aligned} \tag{4.80}$$

Because of the special form of the interaction, it is possible to solve this problem exactly by using methods very similar to those employed in the solution of the normal modes problem. One can easily show that the *exact* eigenvalues of $A_0 + \epsilon A_1$ are given by

$$\lambda_{\nu,\mu} = \hbar\omega_+(\nu + \tfrac{1}{2}) + \hbar\omega_-(\mu + \tfrac{1}{2}) ,$$

where $\omega_\pm = \omega_0\sqrt{1 \pm \epsilon}$. In particular, we get

$$\begin{aligned} \lambda_{0,0} &= \frac{\hbar}{2}(\omega_+ + \omega_-) = \hbar\omega_0(1 - \epsilon^2/8 + \cdots) , \\ \lambda_{1,0} &= \tfrac{3}{2}\hbar\omega_+ + \tfrac{1}{2}\hbar\omega_- = \hbar\omega_0(2 + \tfrac{1}{2}\epsilon - \tfrac{1}{4}\epsilon^2 + \cdots) , \\ \lambda_{0,1} &= \tfrac{1}{2}\hbar\omega_+ + \tfrac{3}{2}\hbar\omega_- = \hbar\omega_0(2 - \tfrac{1}{2}\epsilon - \tfrac{1}{4}\epsilon^2 + \cdots) , \end{aligned}$$

which correspond exactly to our Eqs. (4.78) and (4.80) with a slight change in notation.

Similar calculations apply to the higher levels. The reader may want to examine the triply degenerate states $\phi_{2,0}$, $\phi_{1,1}$, and $\phi_{0,2}$ which all have $\lambda^{(0)} = 3\hbar\omega_0$ (Problem 23). Solutions of the type given above are readily obtained for this case also.

PROBLEMS

1. Prove that if A is self-adjoint, then det A is real. (This is all one can say about the determinant of a self-adjoint matrix, whereas if B is unitary, then $|\det B| = 1$. Prove this also.)
2. Prove that if the matrices A and B are related by a similarity transformation, $A = P^{-1}BP$, and P is unitary, then the similarity transformation preserves Hermiticity.
3. Prove that the $n \times n$ matrix $(I - iA)$ has an inverse if A is Hermitian.
4. Let B be an $n \times n$ matrix. Define the $n \times n$ matrix $A = e^{iB}$ in terms of the power-series expansion of the exponential. That is,

$$A = \sum_{n=0}^{\infty} \frac{(iB)^n}{n!} = I + iB + \frac{(iB)^2}{2!} + \cdots$$

   Assuming that the series converges, prove that A is isometric if B is self-adjoint. Also show that any isometry U can be written as $U = \exp(iH)$, where H is Hermitian. Obtain an expression for H in terms of quantities related to U. [*Hint*: the properties of the eigenvectors and eigenvalues of the operators will be helpful.]
5. Prove that the eigenvalues of any anti-self-adjoint matrix are imaginary.

6. Either prove or find a counterexample to the following statements:
   a) If A and B are $n \times n$ matrices, then $AB = 0$ implies that either $A = 0$ or $B = 0$.
   b) An $n \times n$ Hermitian matrix which is completely degenerate (all eigenvalues equal) is necessarily diagonal.
   c) If an $n \times n$ matrix is Hermitian, then all its powers are Hermitian.
   d) $\det(A + B) = \det A + \det B$.
   e) If A and B are $n \times n$ Hermitian matrices, then AB is Hermitian.
   f) $\operatorname{tr} AB = \operatorname{tr} A \operatorname{tr} B$ in a finite-dimensional space.
7. a) Prove that two vectors x and y in a complex inner-product space are orthogonal if and only if $\|\alpha x + \beta y\|^2 = \|\alpha x\|^2 + \|\beta y\|^2$ for all α and β. Does this result hold if we have just $\|x + y\|^2 = \|x\|^2 + \|y\|^2$?
   b) Prove that in any inner-product space,
   $$\|x - y\| \geq |\ \|x\| - \|y\|\ |.$$
   c) Prove the triangle inequality
   $$\|x + y\| \leq \|x\| + \|y\|.$$
8. In classical mechanics, the virial theorem states that the average value of the potential energy of a simple harmonic oscillator is equal to the average value of its kinetic energy. If $(\phi, A\phi)$ denotes the average value in the state ϕ of an observable A (assuming that ϕ is normalized to unity), show that this result is also true for the quantum-mechanical harmonic oscillator. Show that the polarizability of an electron bound by an harmonic force is the same as is found classically, namely e^2/k, where e is the electric charge and k is the spring constant. This means that the dielectric constant for a collection of such objects (density $= n$ per unit volume) at zero frequency is $4\pi n e^2/k$.
9. Define $\exp(A)$, where A is any matrix, by the power series expansion of $\exp(x) = 1 + x + \cdots$. Show that if there exists a matrix P which reduces A to diagonal form by a similarity transform, then one has
   $$\det \exp(A) = \exp \operatorname{tr}(A).$$
10. Under what circumstances can you prove that AB and BA have the same eigenvalues? If they do have the same eigenvalues, what is the relation between the corresponding eigenvectors?
11. The set of linear transformations on some vector space can itself be considered as a vector space. Show that an inner product for this space is given by $(A, B) = \operatorname{tr}(A^\dagger B)$, where A and B are two linear operators on the space in question, that is, show that this satisfies all the axioms of inner-product spaces.
12. *The Lorentz group.*
    a) Evaluate the determinant of the Lorentz matrix. Comments?
    b) Prove that the set of Lorentz transformations in one velocity direction form a group.
13. *The relativistic law for the sum of two velocities.* Let frame K' move with uniform velocity v parallel to K and let K'' move with uniform velocity v' relative to K'. What is the velocity v'' of K'' relative to K? Prove that $v'' \leqq c$ if v and v' are $\leqq c$. Thus c is an upper limit for the relative velocity of any two frames.
    Suggested method: In proving closure in Problem 12, one finds that the product of one Lorentz transformation (complex orthogonal transformation through

"angle" $\theta = \tanh^{-1}\beta$) and another Lorentz transformation (complex rotation through $\theta' = \tanh^{-1}\beta'$) is again a Lorentz transformation (through $\theta + \theta' = \theta'' = \tanh^{-1}\beta''$). Now solve for β'' (that is v'') in terms of v and v'.

14. Derive the matrix A of the general Lorentz transformation corresponding to relative velocity $v/c = \tanh\alpha$ in a direction given by the spherical angles ϕ, θ by a similarity transformation, $R^{-1}LR$, where R is the extension of the rotation matrix, $R(\phi, \theta)$, of Eq. (1.23) to a 4×4 matrix that acts only on the three space coordinates. Note that $R^{-1} = \tilde{R}$. Assume that the axes of K and K' remain parallel during their uniform relative motion. Check your answer to see whether it reduces properly in the 3 special cases: $\phi = \phi$, $\theta = 0$; $\phi = 0$, $\theta = \pi/2$; $\alpha = 0$.

15. Show that the general Lorentz transformation worked out in Problem 14 may be expressed in vector notation as follows:

$$\mathbf{r}' = \mathbf{r} + \frac{(\mathbf{v}\cdot\mathbf{r})\mathbf{v}}{v^2}(\gamma - 1) - \mathbf{v}t\gamma\,, \qquad t' = \left(t - \frac{\mathbf{v}\cdot\mathbf{r}}{c^2}\right)\gamma\,,$$

where $\gamma = 1/\sqrt{1-\beta^2}$, $\beta = v/c$, and $\mathbf{v}$ is the relative velocity of the two frames. These formulas are a compact, convenient way to write the Lorentz transformation between systems with parallel axes moving with arbitrarily directed uniform relative velocity $\mathbf{v}$.

16. a) Find the eigenvalues and eigenvectors of the Lorentz transformation. Verify that the eigenvectors are null vectors.
    b) Evaluate the determinant of the pure Lorentz matrix.
    c) Any other matrix representation of the Lorentz transformation can be found by performing a similarity transformation on the matrix A (Problem 14). What is the value of the determinant of *any* matrix representation of the Lorentz transformation?

17. The *projection* operator P_i is defined as follows:

$$x' = P_i x \equiv x_i(x_i, x)\,,$$

where x_i is a unit vector. This operator is called a projection operator because all x' are in the direction of x_i and the length of x' equals the component of x in the x_i direction, namely (x_i, x). In the following, let the x_i be a set of orthonormal vectors which span the n-dimensional vector space V. *Prove*:

a) P_i is idempotent.
b) $P_iP_j = 0$ for $i \neq j$. Interpret geometrically.
c) P_i has no inverse.
d) $\sum_{i=1}^{n} P_i = I$.
e) P_i is Hermitian.
f) Let A be a self-adjoint linear operator defined on an n-dimensional vector space V with n-eigenvalues, ϵ_i, and n-eigenvectors, x_i. Prove that A may be written as

$$A = \sum_{i=1}^{n} \epsilon_i P_i\,,$$

where P_i is as defined above. This is known as the spectral theorem for self-adjoint linear operators. (The analogous theorem for normal operators can also be easily proved.) Note the consistency of this result with parts (d) and (e) of this problem.

18. Given any linear transformation A on a finite-dimensional vector space, prove that in a given coordinate system, the adjoint matrix always exists.
[*Hint*: Modify the reasoning of Theorem 4.7.]
19. Obtain an upper bound for the lowest eigenvalue of the quantum harmonic oscillator by taking as a trial vector (function)

$$\phi = c \exp(-\alpha x^2/2) .$$

Note that this is exactly the result derived in Section 3.10.
20. In the text we obtained the first-order correction to the eigenfunctions of a linear operator with a nondegenerate spectrum. Using these methods, obtain the second-order correction. You may assume that the second-order correction is orthogonal to the zeroth order state from which you start.
21. Suppose that the harmonic oscillator Hamiltonian,

$$A_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 , \qquad [x, p] = i\hbar ,$$

is perturbed by a small term $A_1 = \epsilon x$. Find the first and second-order corrections to the eigenvalues of A_0 produced by A_1. Since this problem can be solved exactly, you can check your answers.
22. Find the first few corrections in perturbation theory to the lowest eigenvalue and corresponding eigenvector for the coupled harmonic-oscillator problem discussed in Section 4.12.
23. Using degenerate perturbation theory, find the corrections to the eigenvalues and eigenvectors of the coupled harmonic oscillator problem which correspond to the unperturbed eigenvalues $\lambda^{(0)} = 3\hbar\omega_0$.
24. Define an operator P_n by

$$P_n x = \sum_{i=1}^{\mu_n} x_{n,i}^{(0)} (x_{n,i}^{(0)}, x) ,$$

where $i = 1, \cdots, \mu_n$ is the degree of degeneracy of the nth eigenvalue of some operator A_0, and $x_{n,i}^{(0)}$ are the corresponding eigenvectors. Let ϵA_1 be the perturbation. Show that if we define

$$A_0' \equiv A_0 + \epsilon \sum_n P_n^\dagger A_1 P_n , \qquad A_1' \equiv A_1 - \sum_n P_n^\dagger A_1 P_n ,$$

then it is possible to solve for the eigenfunctions and eigenvalues of A_0' in terms of the eigenfunctions and eigenvalues of A_0. If the degeneracy of A_0 is completely lifted in first order by A_1, then show that we can do ordinary nondegenerate perturbation theory on A_1'. Show that the first-order corrections to the eigenvectors of A_0'. are all zero. Make any comments that seem relevant about the relative sizes of terms in the first few orders of perturbation theory.
25. Consider a four-dimensional real vector space with basis vectors $\{e_0, e_1, e_2, e_3\}$ and define a multiplication of basis vectors as follows:

$$e_0 e_i = e_i e_0 = e_i \qquad (i = 1, 2, 3)$$
$$e_0 e_0 = e_0, \qquad e_i e_i = -e_0 \qquad (i = 1, 2, 3)$$
$$e_i e_j = \sum_{k=1}^{3} \epsilon_{ijk} e_k \qquad (i \neq j;\ i, j = 1, 2, 3)$$

Because of this last equation it is customary to write $e_1 = \mathbf{i}$, $e_2 = \mathbf{j}$ and $e_3 = \mathbf{k}$ and hence to write a general vector in this space as

$$\begin{aligned} q &= q_0e_0 + q_1e_1 + q_2e_2 + q_3e_3 \\ &= q_0e_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k} \\ &= q_0e_0 + \mathbf{q}\,. \end{aligned}$$

(a) Extend the definition of vector multiplication to all vectors in the space by linearity. (Following Sir W. R. Hamilton we call the elements of the space *quaternions*.) Let $q = q_0e_0 + \mathbf{q}$ and $p = p_0e_0 + \mathbf{p}$ be any two quaternions. Show that $pq = (p_0q_0 - \mathbf{p}\cdot\mathbf{q})e_0 + (p_0\mathbf{q} + q_0\mathbf{p} + \mathbf{p}\times\mathbf{q})$.

(b) When does $pq = qp$?

(c) Show that the unit quaternion is equal to e_0, that is, $e_0q = q = qe_0$.

(d) Prove that if $pq = 0$, then either $p = 0$ or $q = 0$.

(e) Show that $(pq)r = p(qr)$. Hence the set of quaternions forms what is called an *algebra*.

(f) Let $q = q_0e_0 + \mathbf{q}$ be a quaternion; define its quaternion *conjugate* to be $q^* = q_0e_0 - \mathbf{q}$. Show, using this definition, that if $q \neq 0$, there exists a unique inverse quaternion, q^{-1}, for which $qq^{-1} = e_0 = q^{-1}q$. This shows that the quaternions form a non-commutative field.

(g) Show that a realization of the quaternion field is obtained by writing

$$e_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\ e_1 = -i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ e_2 = -i\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\ e_3 = -i\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

where $i = \sqrt{-1}$.

26. The quantum mechanical Hamiltonian for a charged particle in a magnetic field is $H = (\mathbf{p} - (e/c)\mathbf{A})^2/2m$, where $\mathbf{A}$ is the vector potential. We assume that the magnetic field is constant in magnitude and direction.

(a) Show that a vector potential for such a field is $\mathbf{A} = \mathbf{B}\times\mathbf{r}/2$, where $\mathbf{B}$ is the constant field, i.e., show that $\nabla\times\mathbf{A} = \mathbf{B}$. Show also that $\nabla\cdot\mathbf{A} = 0$.

(b) Assuming for convenience that the magnetic field is in the z-direction and that the particle's momentum in the z-direction is zero, that is, $p_z = 0$, show that if we take $p_x = (\hbar/i)d/dx$ and $p_y = (\hbar/i)d/dy$ then the Hamiltonian can be written as

$$H = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{1}{2}m\omega^2x^2 + \frac{1}{2}m\omega^2y^2 - \omega L_z\,,$$

where $\mathbf{L} = \mathbf{r}\times\mathbf{p}$ and $\omega = eB/2mc$.

(c) Using the fact that p_x, p_y, x and y are self-adjoint operators, define annihilation operators a_x and a_y by

$$a_x = (2m\hbar\omega)^{-1/2}(xm\omega + ip_x) \text{ and } a_y = (2m\hbar\omega)^{-1/2}(ym\omega + ip_y)$$

so that we have $[a_x, a_x^\dagger] = [a_y, a_y^\dagger] = I$ and $[a_x, a_y] = [a_x, a_y^\dagger] = 0$. Show that H takes the form

$$H = \hbar\omega[a_x^\dagger a_x + a_y^\dagger a_y + 1 + i(a_x^\dagger a_y - a_y^\dagger a_x)]$$

(d) Define $b = (a_x + ia_y)/\sqrt{2}$. Show that

(i) $H = 2\hbar\omega(b^\dagger b + 1/2)$

(ii) $[b, b^\dagger] = I$

(iii) $[b, H] = 2\hbar\omega b$, $[b^\dagger, H] = -2\hbar\omega b^\dagger$.

(e) Now solve the eigenvalue problem $H\phi = E\phi$. Assuming that there is a lowest eigenvalue show that it equals $\hbar\omega$ and that the corresponding eigenvector(s) must satisfy $b\phi_0 = 0$. Show that the remaining (normalized) eigenvectors and eigenvalues are given by

$$\phi_n = (b^\dagger)^n \phi_0 / \sqrt{n!}$$
$$E_n = 2\hbar\omega(n + 1/2) ,$$

or, in a more conventional notation, $E_n = \hbar\omega_L(n + 1/2)$, where $\omega_L = 2\omega = eB/mc$ is the Larmor frequency.

(f) Let u_0 be the unique solution to $a_x u_0 = 0$ and let v_0 be the unique solution to $a_y v_0 = 0$. (See Section 3.10; u_0 and v_0 are simply normalized Gaussians in the x and y variables, respectively.) Then clearly the function $\phi_0^{(0)} = u_0 v_0$ satisfies $b\phi_0^{(0)} = 0$, so we have easily found one solution for the ground state. However, this is not the *only* solution. Define an operator, d, by $d = (a_x - ia_y)/\sqrt{2}$ and show that $[b, d] = 0$ and $[b, d^\dagger] = 0$. Hence show that if $\phi_0^{(0)}$ is defined as above, then

$$\phi_0^{(m)} = (d^\dagger)^m \phi_0^{(0)} / \sqrt{m!}$$

also satisfies $b\phi_0^{(m)} = 0$ for *any* m, and $(\phi_0^{(m)}, \phi_0^{(m')}) = \delta_{mm'}$. Thus, the ground state is *infinite-fold degenerate*, and hence so is each excited state. In fact, we have

$$\phi_n^{(m)} = (b^\dagger)^n \phi_0^{(m)} / \sqrt{n!} = (d^\dagger)^m \phi_n^{(0)} / \sqrt{m!} .$$

27. For a vector space, V, with a pseudo inner product defined by Eqs. (4.23), (4.24) and (4.25) we make the following definitions: For any operator A, we *define* the transpose operator $\tilde{A}$ by $[x, Ay] = [\tilde{A}x, y]$ for all x and y. S is said to be symmetric if $\tilde{S} = S$. O is said to be orthogonal if $\tilde{O}O = I$. Two vectors, $x \neq y$, are said to be *pseudo-orthogonal* if $[x, y] = 0$, and finally a set of *non-null* vectors, x_i, is said to be a complete, pseudo-orthonormal set if the x_i span the space and if $[x_i, x_j] = \delta_{ij}$.

(a) Prove that if $X = \{x_i\}$ is a complete, pseudo-orthonormal set, then for any x in V

(i) $x = \sum_{i=1}^{n} [x_i, x] x_i$

(ii) $[x, y] = \sum_{i=1}^{n} [x, x_i][x_i, y]$.

(b) Show that the matrix of a symmetric operator with respect to a complete, pseudo-orthonormal basis satisfies $S_{ij} = S_{ji}$. Show that the matrix elements of an orthogonal operator O satisfy $\sum_{k=1}^{n} O_{ik} O_{kj} = \delta_{ij}$.

(c) Show that if S is a symmetric operator and $Sx_1 = \lambda_1 x_1$ and $Sx_2 = \lambda_2 x_2$, then $[x_1, x_2] = 0$ for $\lambda_1 \neq \lambda_2$. Show that if all the eigenvalues of S are *distinct*, then the eigenvectors of S form a complete, pseudo-orthonormal set. [Hint: you must exclude the possibility of a null-eigenvector.]

(d) Give an example of a symmetric 2×2 matrix whose eigenvalues are *not* real, but whose eigenvectors are nevertheless pseudo-orthogonal.

(e) Show that every eigenvalue of an orthogonal operator belonging to a *non-null* eigenvector is either 1 or -1.

(f) Prove that if O is an orthogonal transformation and $Ox_1 = \lambda_1 x_1$ and $Ox_2 = \lambda_2 x_2$, then $[x_1, x_2] = 0$ if $\lambda_1 \neq \lambda_2^{-1}$. Show further that if either x_1 or x_2 is non-null, then $[x_1, x_2] = 0$ if $\lambda_1 \neq \lambda_2$.

FURTHER READINGS

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CHAPTER 5

HILBERT SPACE—COMPLETE ORTHONORMAL SETS OF FUNCTIONS

INTRODUCTION

The principal strands of mathematics—algebra, geometry, and analysis—are joined when functions are viewed as vectors in a vector space. This unification is the subject of this chapter; none of the results or methods of mathematics has more relevance to modern physics.

Throughout this chapter we shall be using concepts, terminology, and theorems from Chapters 3 and 4. The present chapter represents the infinite—dimensional generalization of certain of the results obtained there. However, the fact that the vectors are now functions means that there are a number of additional considerations and possibilities in the development of the theory. These result primarily from the attempt to represent a function as a linear combination of some given set of functions—i. e., the problem of series expansions. All the characteristic questions of analysis, such as those of convergence, therefore become relevant.

In particular, the so-called "special functions" of mathematical physics—spherical harmonics, Legendre, Hermite, and Laguerre polynomials, to name a few—are all conveniently treated within the framework of Hilbert space. As we deal with the various special functions, we shall discuss the differential equations of which they are solutions. But the treatment will not be based on the differential equations. The framework of Hilbert space is a much more comprehensive one, and unifies what is otherwise a bewildering maze of special cases and properties.

There will be three stages in the discussion of these functions. In the first stage, where we introduce the basic notions of Hilbert space, the treatment will be quite abstract; the special functions will be mentioned only in passing. Then we shall consider the special functions one by one, somewhat inductively, de-emphasizing their common origins and properties. Finally, we shall return to a more abstract level and systematize the properties of these functions in a single comprehensive framework.

Hilbert space is the mathematical setting for quantum physics. Physical observables are represented by operators in Hilbert space, and physical states are

vectors (functions) in Hilbert space. There is no more important property of the functions that describe the possible states of a physical system than that they form a *complete* set. The whole of quantum theory is based on this fact. Thus most of this chapter will in one way or another have to do with the completeness of sets of functions.

At the end of the chapter we shall bring together the key results of the theory of Hilbert space that have been developed throughout the chapter, and emphasize their role in the formulation of quantum mechanics.

5.1 FUNCTION SPACE AND HILBERT SPACE

In discussing finite-dimensional vector spaces, we referred several times to P_n, the vector space of polynomials of degree $\leq n$. The vectors in that vector space are a class of simple functions, the polynomials.

We are now going to define another vector space whose elements are functions. The elements of this space are the complex-valued functions of a real variable x, defined on the closed interval $[a, b]$,* which are *square integrable*.† We shall show that the set of square integrable functions forms a vector space. This space is called L_2 by mathematicians: we shall call it *function space*. It will be found to be an infinite-dimensional space.

Intuitively, it would seem that function space is far "larger" than the finite-dimensional space P_n, and in a sense this is so. However, suppose that the basis functions for P_n—the set $\{x^m, m = 0, 1, \cdots, n\}$—were extended to contain all possible powers of x by letting $n \to \infty$. Weierstrass's theorem, the central result of this chapter, shows that this infinite set of functions is not as poverty-stricken as it might appear. We shall shortly give a precise account of its latent powers. First we return to the definition of function space.

Addition of the two vectors, f_1 and f_2, in function space is defined according to the natural rule:

$$(f_1 + f_2)(x) \equiv f_1(x) + f_2(x) ,$$

and multiplication by a complex scalar α is defined as

$$(\alpha f)(x) \equiv \alpha f(x) .$$

The only possible difficulty in showing that these operations satisfy the various axioms that define a vector space is establishing closure. That is, are the sums and scalar multiples of square-integrable functions also square-integrable? The

* A *closed interval*, written $[a, b]$, is the set of all points $\{x\}$ such that $a \leq x \leq b$. An *open interval*, which we write as (a, b), is the set of points $\{x\}$ such that $a < x < b$. A closed interval is always finite; if either a or b is infinite, the interval is open at that end. Later in the chapter, when we deal with infinite intervals, we shall alert the reader by explicit mention of the shift from finite to infinite intervals.

† A function is *square integrable* on $[a, b]$ if $\int_a^b |f(x)|^2 dx$ exists and is finite.

answer is yes, and so the space is in fact a vector space. We prove closure of the sum.

$$\begin{aligned} |f_1 + f_2|^2 &= |f_1|^2 + |f_2|^2 + f_1^* f_2 + f_1 f_2^* \\ &= |f_1|^2 + |f_2|^2 + 2\mathrm{Re}\,(f_1^* f_2) \\ &\leqq |f_1|^2 + |f_2|^2 + 2|f_1^* f_2| \\ &\leqq |f_1|^2 + |f_2|^2 + 2|f_1||f_2| \,. \end{aligned} \tag{5.1}$$

Also,

$$0 \leqq (|f_1| - |f_2|)^2 = |f_1|^2 + |f_2|^2 - 2|f_1||f_2| \,,$$

so

$$|f_1|^2 + |f_2|^2 \geqq 2|f_1||f_2| \,.$$

We use this last inequality to replace $2|f_1||f_2|$ in Eq. (5.1) with something larger, thereby preserving the inequality. Thus the inequality

$$|f_1 + f_2|^2 \leqq 2|f_1|^2 + 2|f_2|^2 \,, \tag{5.2}$$

holds at every point in $[a, b]$. Integrating over both sides, we see that square integrability of f_1 and f_2 ensures square integrability of their sum.

We now add an inner product to function space.

Definition 5.1. The inner product of two functions, f_1 and f_2, belonging to function space, is defined by

$$(f_1, f_2) \equiv \int_a^b f_1^*(x) f_2(x)\, dx \,. \tag{5.3}$$

Note that square integrability implies that $(f, f) \equiv ||f||^2 = \int_a^b |f|^2\, dx < \infty$. The quantity $||f||$ is called the norm of f. The inner product for any pair of square-integrable functions also exists. Since

$$|f_1^* f_2| = |f_1||f_2| \leqq \tfrac{1}{2}(|f_1|^2 + |f_2|^2) \,,$$

it follows that

$$\int_a^b |f_1^* f_2|\, dx \leqq \tfrac{1}{2}(||f_1||^2 + ||f_2||^2) < \infty \,;$$

but any function whose absolute value is integrable is itself integrable, so (f_1, f_2) exists.

Verification that (f_1, f_2) as defined in Eq. (5.3) is an inner product proceeds very smoothly until we come to the question of positive-definiteness. Clearly,

$$(f, f) \equiv ||f||^2 = \int_a^b |f|^2\, dx \geqq 0 \,.$$

But does $(f, f) = 0$ imply that $f(x) = 0$ for *all* x in $[a, b]$? Not quite, for the function $f(x)$ can be nonzero at any finite number of points, and the integral will not "notice" this. That is, there will be no contribution to the integral even though the integrand is not identically zero throughout $[a, b]$.

The discussion of situations like these is facilitated if a slightly more general notion of integration is used. The Riemann integral suffers from several difficulties; for example, let us consider an extreme case by looking at the bizarre function $f(x)$ defined to be 1 for every rational number in [0, 1] and 0 for every non-rational number. Now, since there are "very few" rationals, only a countable number in fact, we strongly suspect that the integral of this function is zero. However, if we form the upper and lower Riemann integrals by partitioning [0, 1] into small segments Δx_i, and write

$$\overline{\int} f(x)\,dx = \sum_i \Delta x_i \max\,[f(x)], \qquad x_i \leq x \leq x_i + \Delta x_i$$

$$\underline{\int} f(x)\,dx = \sum_i \Delta x_i \min\,[f(x)], \qquad x_i \leq x \leq x_i + \Delta x_i$$

in the usual way, we see that no matter how small the subinterval, Δx_i, the maximum of $f(x)$ on this interval is always 1 and the minimum is always zero. Thus

$$\overline{\int} f(x)\,dx = 1 \qquad \text{and} \qquad \underline{\int} f(x)\,dx = 0\,,$$

so the Riemann integral does not exist.

In the theory of Lebesgue integration, this integral does exist and it equals zero. We say that $f(x) = 0$ except on a set of points of *measure zero*, or $f(x) = 0$ *almost everywhere*. The intuitive content of this phrase is simple. If we have a countable number of points on the real line and are given a small strip of paper of length ϵ, then we can paste a small piece of the strip over each element of the set by dividing it into a countable number of pieces of width $\epsilon/2^n$. Since $\sum_{n=1}^{\infty} \epsilon/2^n = \epsilon$, we use up only our given strip in the process. But since the original strip can be arbitrarily small, the set of points on which $f(x)$ is nonzero is negligible with respect to the set on which it is zero, despite the fact that every real number is arbitrarily close to some rational. Thus, the rationals are a set of measure zero on the real line.

We shall go no further with the notion of the Lebesgue integral, since all the functions which we will actually want to integrate in this book will be Riemann integrable. For example, the Riemann integral always exists for the class of piecewise continuous functions defined on closed intervals. In any case, the Riemann integral, when it exists, is equal to the Lebesgue integral.

To summarize: If f is a function in function space and $(f, f) = 0$, then $f(x)$ need not be identically zero for all x, but it can differ from zero only on a set of measure zero. We say that $(f, f) = 0$ implies $f(x) = 0$ almost everywhere. Any function which equals zero almost everywhere is called the zero function. With this broadened concept of the zero function, Axiom 3 for an inner product is satisfied, so Definition 5.1 does define an inner product.

Thus we have a complex inner product defined in function space, which is itself a complex vector space of complex square-integrable functions of a *real* variable ranging through the closed interval $[a, b]$.

To be useful to the physicist, the inner-product space must also be *complete*. A complete space is one in which there exists no Cauchy sequence of elements of the space which tend toward limits outside the space.* An elementary example of an incomplete space is the set of rational numbers. The sequence of partial sums, $S_N = \sum_{n=0}^{N} 1/n!$ is a sequence of rationals, but it converges to the number e, which is irrational. However, a basic theorem of real analysis states that the set of all real numbers is complete; one cannot get out of the set by taking the limits of Cauchy sequences of real numbers.

Similarly, we want to discover a class of *functions* having the property that there will be no Cauchy sequence of functions in this class whose limits do not belong to the class. Such a class of functions is *complete*. This absolutely fundamental problem in analysis is solved by the Riesz-Fischer theorem, which states that the space of square-integrable functions, i. e., functions with a finite norm, is complete. The proof may be found in any book on functional analysis (e. g., Riesz and Nagy, Rudin). The theorem may be stated as follows.

Riesz-Fischer Theorem. Let the functions $f_1(x), f_2(x), \cdots$ be elements in function space. If

$$\lim_{n,\, m\to\infty} ||f_n - f_m||^2 \equiv \lim_{n,\, m\to\infty} \int_a^b |f_n(x) - f_m(x)|^2\, dx = 0\,,$$

then there exists a square (Lebesgue) integrable function $f(x)$ to which the sequence $f_n(x)$ converges "in the mean"; i. e., there exists an f such that

$$\lim_{n\to\infty} \int_a^b |f(x) - f_n(x)|^2\, dx = 0\,.$$

We shall discuss convergence in the mean in greater detail in the following section. It should be emphasized that this theorem is not true unless the integral used is the Lebesgue integral.

Thus function space, as we have defined it, is in fact complete. Henceforth we shall call this complete inner-product space by its usual name, *Hilbert space*, although function space is not the only Hilbert space.

The notions of orthogonality, normalization, and orthonormal sets of functions are defined exactly as for vectors. Thus the set of functions $\{f_i\}$ is said to be *orthonormal* if

$$(f_i, f_j) \equiv \int_a^b f_i^*(x) f_j(x)\, dx = \delta_{ij}\,.$$

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* A *Cauchy sequence* is a sequence $\{s_n\}$ which has the property that given an arbitrary number $\epsilon > 0$, there is an index $N(\epsilon)$ such that if m and n are larger than $N(\epsilon)$,

$$||s_n - s_m|| < \epsilon.$$

If the sequence in question is a sequence of numbers, then the norm is interpreted as the absolute value. If, as in our case, the sequence is a sequence of functions, then the norm is given by

$$||f|| \equiv \left[\int_a^b |f|^2\, dx\right]^{1/2}.$$

Example. *The Fourier functions.* Let $f_n(x) = e^{inx}/\sqrt{2\pi}$, where $n = 0, \pm 1, \pm 2, \cdots$ The set $\{f_n\}$ is orthonormal over the interval $[-\pi, \pi]$.

Proof.

$$(f_n, f_m) = \int_{-\pi}^{+\pi} f_n^* f_m \, dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(m-n)x} \, dx$$

$$= \frac{1}{2\pi(m-n)i} e^{i(m-n)x} \Big]_{-\pi}^{\pi} = 0. \qquad \text{if } m \neq n \, ;$$

$$(f_n, f_n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 \, dx = 1, \qquad \text{if } m = n \, .$$

Therefore

$$(f_n, f_m) = \delta_{nm} \, .$$

An often used generalization of orthonormality is the following:

Definition 5.2. A set of functions $\{f_n\}$ is orthonormal with respect to a nonnegative weight function $w(x)$ on $[a, b]$ if

$$(f_n, f_m) \equiv \int_a^b f_n^*(x) f_m(x) w(x) \, dx = \delta_{nm} \, .$$

We shall work almost exclusively with orthonormal sets of functions, because, as in finite-dimensional vector spaces, they greatly simplify computations.

5.2 COMPLETE ORTHONORMAL SETS OF FUNCTIONS

In the theory of finite-dimensional vector spaces, we found a number of equivalent alternative characterizations of a complete set of basis vectors (Theorem 4.3). The corresponding problem in Hilbert space is that of representing a function as a linear combination of some given set of functions, or, in other words, the problem of series expansions of functions in terms of a given set. The prototype of all such series expansions is the Fourier series. Our treatment, however, will be a general one, embracing many of the functions of mathematical physics; Fourier series will be treated within this framework as a special case.

The first question we must face is that of defining the *completeness of an orthonormal set of functions* in Hilbert space. (Completeness of a *set* of functions is not the same as completeness of a *space*, mentioned briefly in the last section; however, they are intimately connected as we shall see.) We could say that an orthonormal set of functions $\{f_i(x)\}$ is complete if any function $f(x)$ in Hilbert space is expressible as a linear combination of the $f_i(x)$:

$$f(x) = \sum_{i=1}^{\infty} c_i f_i(x) \, ,$$

the series converging to f at every point x. This would provide a close analog of the idea of completeness of a set of basis vectors in finite-dimensional spaces. But this criterion of convergence is, for many purposes, unnecessarily severe and exclusive. As we shall see, by this criterion there would exist no complete

orthonormal set of functions in Hilbert space. So, instead of demanding strict pointwise convergence, we shall weaken the convergence criterion, and in this way permit the existence of complete sets of functions.

The appropriate weakening of the convergence criterion is suggested by the slight difficulty we encountered with the positive-definiteness of the inner product. There we found that

$$(f, f) = \int_a^b |f|^2 \, dx = 0$$

implied that $f(x)$ vanished not at *every* point x in $[a, b]$, but rather, almost everywhere in $[a, b]$; that is, at all but a set of points of measure zero. Similarly, we shall say that $\sum_i c_i f_i(x)$ converges "in the mean" to $f(x)$ if

$$\lim_{n\to\infty} \int_a^b |f(x) - \sum_i^n c_i f_i(x)|^2 \, dx = 0 .$$

This allows the series to differ from $f(x)$ on a set of measure zero.

In order to develop the notion of completeness of an orthonormal set of functions further, we need to define the different kinds of convergence we shall be using:

1. Pointwise convergence;
2. Uniform convergence;
3. Convergence in the mean.

Definition 5.3. A sequence of functions, $h_n(x)$, converges pointwise to $h(x)$ on $[a, b]$ if for every x in $[a, b]$ and every $\epsilon > 0$ there exists an integer $N(x, \epsilon)$ such that for $n > N$,

$$|h(x) - h_n(x)| < \epsilon .$$

The $h_n(x)$ may themselves be the partial sums of another sequence; that is,

$$h_n(x) = \sum_{i=1}^n k_i(x) .$$

The definition of pointwise convergence, and the other types of convergence to follow, may all be stated equivalently in terms of the infinite series $\sum_{i=1}^\infty k_i(x)$, which is the limit of the sequence $h_n(x)$ of partial sums. We say that this series *converges pointwise* to $h(x)$ on $[a, b]$ if for every x in $[a, b]$ and every $\epsilon > 0$, there exists an integer $N(x, \epsilon)$ such that for $n > N$,

$$|h(x) - h_n(x)| = |h(x) - \sum_{i=1}^n k_i(x)| < \epsilon .$$

If there is a single N that works in the above definition for all x in $[a, b]$, the convergence is said to be *uniform*. Formally, we have:

Definition 5.4. A sequence of functions, $h_n(x)$, converges *uniformly* to $h(x)$ on $[a, b]$ if for every $\epsilon > 0$ there exists an integer $N(\epsilon)$, *independent of* x,

such that for $n > N$, $|h(x) - h_n(x)| < \epsilon$, for all x in $[a, b]$. Clearly, uniform convergence implies pointwise convergence.

Note that this definition makes explicit mention of the limit $h(x)$ of the sequence of functions, $h_n(x)$. The *Cauchy criterion* for uniform convergence supplies us with a useful alternative to this definition, in which knowledge of the limit function is *not* assumed. Cauchy's criterion is proved in analysis courses; we state it here.

Theorem 5.1. The sequence of functions $h_n(x)$ converges uniformly on $[a, b]$ if for every $\epsilon > 0$ there exists an integer $N(\epsilon)$ such that for all $r > N$, $s > N$, and x in $[a, b]$, $|h_r(x) - h_s(x)| < \epsilon$.
In terms of the partial sums, $h_n(x) \equiv \sum_{i=1}^{n} k_i(x)$, this becomes

$$|h_r(x) - h_s(x)| = |\sum_{i=1}^{r} k_i - \sum_{i=1}^{s} k_i| = |\sum_{i=r+1}^{s} k_i(x)| < \epsilon .$$

If we have uniform or pointwise convergence, we may write

$$h(x) = \lim_{n\to\infty} h_n(x) = \sum_{i=1}^{\infty} k_i(x) .$$

The weaker convergence we discussed earlier can now be formalized precisely.

Definition 5.5. A sequence of functions $h_n(x)$, converges *in the mean* to $h(x)$ on $[a, b]$ if

$$\lim_{n\to\infty} \int_a^b |h(x) - h_n(x)|^2 \, dx = 0,$$

that is, if for every ϵ there exists an $N(\epsilon)$ such that for $n > N$,

$$\int_a^b |h(x) - h_n(x)|^2 \, dx < \epsilon .$$

The series $\sum_{i=1}^{\infty} k_i(x)$ converges in the mean to $h(x)$ if

$$\lim_{n\to\infty} \int_a^b |h(x) - \sum_{i=1}^{n} k_i(x)|^2 \, dx = 0 .$$

It is easy to see that uniform convergence implies mean convergence. For if convergence is uniform, then for any ϵ there exists an N such that for $n > N$, $|h - h_n| < \epsilon$ for all x in $[a, b]$. Therefore

$$\int_a^b |h - h_n|^2 \, dx < \int_a^b \epsilon^2 \, dx = \epsilon^2(b - a) ,$$

so the integral can be made arbitrarily small by choice of ϵ. This is just the statement that

$$\lim_{n\to\infty} \int_a^b |h - h_n|^2 \, dx = 0,$$

and so we have mean convergence. Note, however, that pointwise convergence does not imply mean convergence (Problem 19).

It is in terms of *mean* convergence that the completeness of an orthonormal set of functions is defined.

Definition 5.6. Let $g(x)$ be any function in Hilbert space (i. e., any square-integrable function) and let $\{f_i(x)\}$ be an orthonormal set of functions in Hilbert space. If there exist constant coefficients $\{a_i\}$ such that the sequence of partial sums $g_n(x) \equiv \sum_{i=1}^{n} a_i f_i(x)$ converges in the mean to $g(x)$, then the *set of functions* $\{f_i\}$ *is a complete orthonormal set.* Equivalently, if the mean square error can be made arbitrarily small,

$$\lim_{n\to\infty} \int_a^b |g - g_n|^2\, dx = \lim_{n\to\infty} \int_a^b |g - \sum_{i=1}^{n} a_i f_i|^2\, dx = 0 ,$$

then the set $\{f_i\}$ is a *complete orthonormal set of functions.* It should be noted that the coefficients $\{a_i\}$ are independent of n. Thus as n increases and we include more terms in the partial sum approximating g, the earlier coefficients do not change. We may say that as we extend the sum to infinity, the infinite *series*

$$\sum_{i=1}^{\infty} a_i f_i$$

approximates the arbitrary function g in the mean. We shall write this as

$$g(x) \doteq \sum_{i=1}^{\infty} a_i f_i(x) .$$

The dot over the equal sign serves to distinguish mean convergence from pointwise convergence.

Since mean convergence does not necessarily imply pointwise or uniform convergence, it should be clear that the completeness of an orthonormal set of functions $\{f_i\}$, expressed by the relation

$$\lim_{n\to\infty} \int_a^b |f - \sum_{i=1}^{n} a_i f_i|^2\, dx = 0 ,$$

or symbolically,

$$f(x) \doteq \sum_{i=1}^{\infty} a_i f_i(x) ,$$

does not imply that

$$f(x) = \sum_{i=1}^{\infty} a_i f_i(x) . \tag{5.4}$$

We may only *equate* $f(x)$ and the expansion series if the series converges pointwise or uniformly to $f(x)$.

Let $f(x)$ be an arbitrary function in Hilbert space, and assume for the moment that we have an orthonormal set of functions $\{f_i(x)\}$ such that the series

$\sum_{i=1}^{\infty} c_i f_i(x)$ converges uniformly to $f(x)$:

$$f(x) = \sum_{i=1}^{\infty} c_i f_i(x) .$$

The coefficients c_i are called the *generalized Fourier coefficients*, or *expansion coefficients*. The formula for them is especially simple because $\{f_i\}$ is an orthonormal set:

$$(f_n, f) = \sum_{i=1}^{\infty} c_i (f_n, f_i) = \sum_{i=1}^{\infty} c_i \delta_{ni} = c_n . \tag{5.5}$$

Since the convergence is uniform, mean convergence is also implied. Therefore

$$\lim_{n\to\infty} \int_a^b |f - \sum_{i=1}^{n} c_i f_i|^2 \, dx = 0 ,$$

[where the c_i are given by Eq. (5.5)], and, consequently, the set of functions $\{f_i\}$ is complete.

Now consider the nonnegative quantity

$$M_n = \int_a^b |f(x) - \sum_{i=1}^{n} a_i f_i(x)|^2 \, dx \geq 0 , \tag{5.6}$$

where $\{f_i\}$ is an orthonormal set, and f is any function in Hilbert space. We want to know: What values of the coefficients a_i will minimize M_n? Or, to put it in the language of the physicist: What values of the a_i will give the best least-squares fit to the arbitrary function $f(x)$?

To answer this question, we expand Eq. (5.6) to obtain

$$\begin{aligned} M_n &= \int_a^b (f^*f - \sum_{i=1}^{n} a_i f^* f_i - \sum_{i=1}^{n} a_i^* f f_i^* + \sum_{i,j=1}^{n} a_i^* a_j f_i^* f_j) \, dx \\ &= (f, f) - \sum_{i=1}^{n} a_i c_i^* - \sum_{i=1}^{n} a_i^* c_i + \sum_{i,j=1}^{n} a_i^* a_j \delta_{ij} , \end{aligned}$$

where $c_i \equiv (f_i, f)$. Adding and subtracting the quantity $\sum_{i=1}^{n} |c_i|^2$, we have

$$M_n = (f, f) + \sum_{i=1}^{n} |a_i - c_i|^2 - \sum_{i=1}^{n} |c_i|^2 \geq 0 .$$

It is clear that M_n is minimized by choosing $a_i = c_i$. We then have

$$\begin{aligned} M_n &= \int_a^b |f(x) - \sum_{i=1}^{n} c_i f_i(x)|^2 \, dx \qquad (5.7) \\ &= (f, f) - \sum_{i=1}^{n} |c_i|^2 \geq 0 , \end{aligned}$$

which may be rewritten as

$$(f, f) \geq \sum_{i=1}^{n} |c_i|^2 = \sum_{i=1}^{n} |(f_i, f)|^2 . \tag{5.8}$$

Writing $s_n = \sum_{i=1}^{n} |c_i|^2$, we see that the sequence $s_1, s_2, \cdots, s_n, \cdots$ is monotonically increasing and bounded above by (f, f), (Eq. 5.8). Therefore s_n tends toward a limit as n becomes arbitrarily large, and we can write

$$(f, f) \geq \sum_{i=1}^{\infty} |c_i|^2 . \tag{5.9}$$

Hence the infinite series $\sum_{i=1}^{\infty} |c_i|^2$ converges. Equation (5.9) is just Bessel's inequality (compare Theorem 4.2) in an infinite-dimensional space.

The orthonormal set $\{f_i\}$ is complete if and only if there exists a set $\{a_i\}$ such that $\lim_{n\to\infty} M_n = 0$. If the set $\{f_i\}$ is complete, then $a_i = c_i$ and the equal sign holds in Bessel's inequality:

$$(f, f) = \sum_{i=1}^{\infty} |c_i|^2 = \sum_{i=1}^{\infty} |(f_i, f)|^2 , \tag{5.10}$$

for every f. This was also true in the finite-dimensional case (see Theorem 4.3). Equation (5.10) is called the *completeness relation*. As in the finite-dimensional case, it can also be stated in the form of Parseval's equation:

$$(f, g) = \sum_{i=1}^{\infty} (f, f_i)(f_i, g) . \tag{5.11}$$

We leave the proof to the reader (Problem 14).

Another characterization of a complete set of n vectors is that there exists no nonzero vector orthogonal to all n vectors in the set. The corresponding statement holds for a complete orthonormal set of functions. We shall present it as a therorem. It will prove very useful in what follows. To facilitate the statement of the theorem, we first make a definition.

Definition 5.7. A set of orthonormal functions is said to be *closed* if no nonzero function is orthogonal to every function in the set.

Theorem 5.2. A set of orthonormal functions in Hilbert space is complete if and only if it is closed.

Proof. We first prove that completeness of the set implies that the set is closed. Assume that there is a nonzero function $f(x)$ (and let it be normalized), such that

$$(f_i, f) \equiv c_i = \int_a^b f_i^*(x) f(x)\, dx = 0$$

for all i. Then

$$\lim_{n\to\infty} \int_a^b |f - \sum_{i=1}^{n} c_i f_i|^2\, dx = \int_a^b |f|^2\, dx = 1 \neq 0 ,$$

(since f is normalized), so the set $\{f_i\}$ is not complete. Thus completeness of an orthonormal set of functions implies that there are no functions that are orthogonal to every member of the set.

We now prove the converse: if the orthonormal set is closed, it is complete. If it is not complete, then the completeness relation, Eq. (5.10), is not satisfied. Thus there exists some function $f(x)$ such that

$$||f||^2 > \sum_{n=1}^{\infty} |c_n|^2 ,$$

where $c_n = (f_n, f)$. But since the above infinite series is convergent, the sequence $\{g_m(x)\}$, where

$$g_m(x) = \sum_{n=1}^{m} c_n f_n(x)$$

is a Cauchy sequence in Hilbert space, and therefore, because of the *completeness of the space*, the $g_m(x)$ must converge in the mean to a limit *in the space*, call it $g(x)$, such that $c_n = (f_n, g)$. Therefore $(f_n, g) = (f_n, f)$ so $(f_n, f\text{-}g) = 0$. Thus $f(x)\text{-}g(x)$ is orthogonal to $f_n(x)$ for all n. We now show that the norm of $f(x)\text{-}g(x)$ is not equal to zero, so $\{f_n(x)\}$ is not closed, contrary to our assumption. It will then follow by contradiction that the set $\{f_n(x)\}$ is complete and the proof will be finished.

Using the inequality

$$||x - y|| \geq |\, ||x|| - ||y||\, |$$

(see Problem 4.7b), we have

$$||f - g|| = ||f - g_m - (g - g_m)|| \geq \Big| ||f - g_m|| - ||g - g_m|| \Big|$$

for all m. Now as $m \to \infty$, we know that $||g - g_m|| \to 0$, whereas

$$||f - g_m||^2 = ||f - \sum_{n=1}^{m} c_n f_n||^2 = ||f||^2 - \sum_{n=1}^{m} |c_n|^2 > 0$$

for all m by assumption. Thus $||f - g|| > 0$ and the proof is complete.

This theorem plays a key role in establishing the completeness of the various orthonormal sets of functions which we shall treat in this chapter. Note the crucial use that is made of the completeness of the Hilbert space in proving it.

We conclude this section with some observations on the uniqueness of the representations of functions in series expansions. We first prove that a function in Hilbert space is uniquely determined almost everywhere by its expansion coefficients with respect to a given complete orthonormal set of functions $\{f_i\}$. Suppose that two functions, f and g, have the same expansion coefficients, that is,

$$c_i = (f_i, f) = (f_i, g) .$$

Thus, $(f_i, f - g) = 0$, so by Theorem 5.2, $f - g = 0$, and hence $f = g$.

Now consider the converse problem. Does a given function have a unique set of expansion coefficients? Assume that

$$\lim_{n\to\infty} ||f - \sum_{i=1}^{n} c_i f_i|| = \lim_{n\to\infty} ||f - \sum_{i=1}^{n} d_i f_i|| = 0 ;$$

that is, assume that there are two partial sums with different expansion coefficients that converge in the mean to the same function f. If the expansion coefficients are unique, then $c_i = d_i$. To prove this, we observe that

$$\begin{aligned}\|\sum_{i=1}^{n} c_i f_i - \sum_{i=1}^{n} d_i f_i\| &= \|\sum_i c_i f_i - f + f - \sum_i d_i f_i\| \\ &\leq \|f - \sum_i c_i f_i\| + \|f - \sum_i d_i f_i\| ,\end{aligned}$$

where we have used the triangle inequality (see Problem 4.7c). Now given any ϵ, we can choose by assumption an n large enough so that both the last two norms are less than $\epsilon/2$. Therefore, for such an n,

$$\|\sum_i c_i f_i - \sum d_i f_i\| = \|\sum_i (c_i - d_i) f_i\| = \left[\sum_i (c_i - d_i)^2\right]^{1/2} < \epsilon .$$

But this can only be true if $c_i = d_i$. Therefore the expansion coefficients of a given function are unique. Since the set $\{f_i\}$ is a complete orthonormal set of functions, it follows from our earlier remarks that $c_i = d_i = (f_i, f)$, the Fourier coefficients.

5.3 THE DIRAC δ-FUNCTION

In this section we introduce Dirac's δ-function in an informal way. The δ-function will play an important role in this book as it usually does in physics.

The first thing to understand about the δ-function is that it is not a function at all. A function is a rule that assigns another number to each number in a set of numbers. The δ-function, as used in physics, is instead a shorthand notation for a rather complicated limiting process whose use greatly simplifies calculations. It takes on a meaning only when it appears under an integral sign, in which case it has the following effect:

$$\int_{-\infty}^{\infty} f(x)\delta(x)\, dx = f(0) . \tag{5.12}$$

A special case of this is $f(x) = 1$. Then

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1 . \tag{5.13}$$

If the singular point is located at an arbitrary point x, then

$$\int_{-\infty}^{\infty} f(x')\delta(x' - x)\, dx' = f(x) . \tag{5.14}$$

Except at the singular point $x = 0$,

$$\delta(x) = 0 . \tag{5.15}$$

Thus $\delta(x)$ behaves as an ordinary function almost everywhere. It vanishes at all points where its argument is not zero, but at that one point it is undefined. Nevertheless its behavior near this point is all that matters.

Now the integral of any real function which vanishes everywhere except at one point must be zero, regardless of the value of the function at the singular point. Thus no *function* which satisfies Eq. (5.15) could possibly satisfy Eqs. (5.12) or (5.13). These equations must be interpreted as a symbolic notation for a process of the following type.

Let $\delta_\alpha(x)$ be a set of functions parametrized by the index α, which have the properties

$$\begin{aligned} &\lim_{\alpha\to 0} \delta_\alpha(x) = 0 \qquad \text{for all } x \neq 0 \,, \\ &\lim_{\alpha\to 0} \int_{-\infty}^{+\infty} f(x)\delta_\alpha(x)\, dx = f(0) \,. \end{aligned} \tag{5.16}$$

Our earlier equations result if we denote "$\lim_{\alpha\to 0} \delta_\alpha(x)$" by $\delta(x)$, and interchange the order of the limiting process and the integration, a procedure which is not, in general, valid. The original equations defining the δ-function must be interpreted as standing for the limiting processes of Eqs. (5.16).

Let us look at several sets of functions that have the properties described in Eqs. (5.16).

1. The simplest possible set of functions which has the proper limiting behavior is depicted in Fig. 5.1(a). The function $\delta_c(x)$ is defined (for $c>0$) by

$$\delta_c(x) \equiv \begin{cases} \dfrac{1}{c} & \text{for } |x| \leq \dfrac{c}{2} \,, \\ 0 & \text{for } |x| > \dfrac{c}{2} \,. \end{cases} \tag{5.17}$$

Clearly, $\lim_{c\to 0} \delta_c(x) = 0$, at all $x \neq 0$. Also, $\int_{-\infty}^{+\infty} \delta_c(x)\, dx = 1$, independent of c. The function $\delta_c(x)$ (and it *is* a function) is defined for all $c \neq 0$, and the limit

$$\lim_{c\to 0} \int_{-\infty}^{\infty} \delta_c(x)\, dx \tag{5.18}$$

is defined and equals 1. Also,

$$\lim_{c\to 0} \int_{-\infty}^{\infty} f(x)\delta_c(x)\, dx = f(0) \,, \tag{5.19}$$

which may be shown formally for continuous functions $f(x)$ as follows:

$$\lim_{c\to 0} \int_{-\infty}^{\infty} f(x)\delta_c(x)\, dx = \lim_{c\to 0} \int_{-c/2}^{c/2} f(x)\delta_c(x)\, dx = \lim_{c\to 0} \frac{1}{c} \int_{-c/2}^{c/2} f(x)\, dx \,;$$

now, by the mean value theorem of integral calculus,

$$\int_{-c/2}^{c/2} f(x) dx = f(\xi c) \int_{-c/2}^{c/2} dx = c f(\xi c) \,,$$

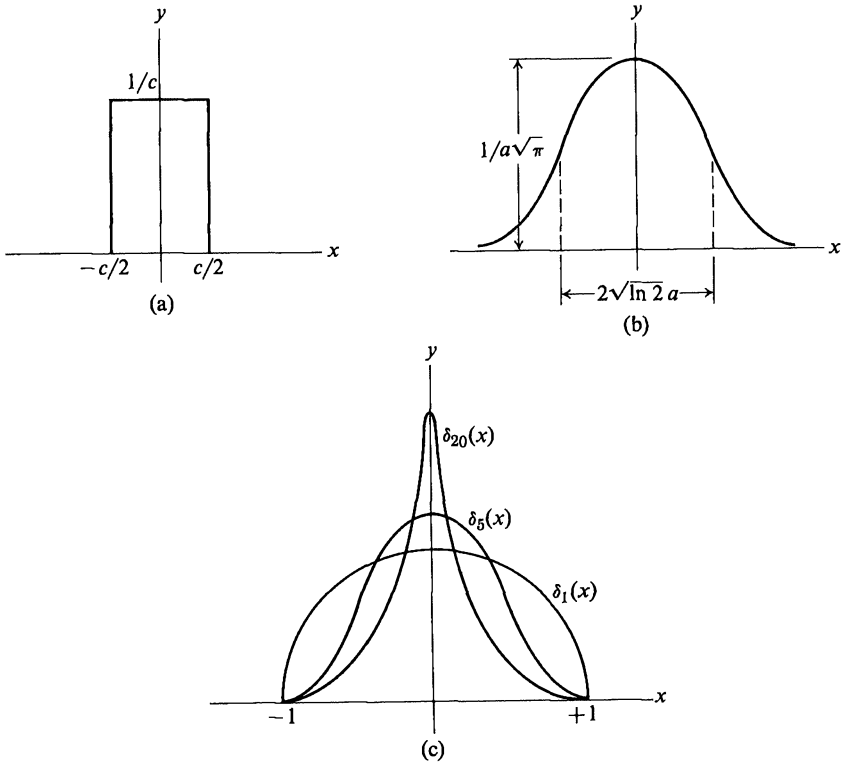

Fig. 5.1. Representations of the δ-function.

where $-1/2 < \xi < 1/2$. Letting $c \to 0$, we obtain

$$\lim_{c\to 0} \int_{-\infty}^{\infty} f(x)\delta_c(x)\, dx = f(0) .$$

2. The sequence of Gaussian distribution functions

$$\delta_a(x) \equiv \frac{1}{a\sqrt{\pi}} e^{-x^2/a^2}$$

provides another representation of the δ-function (see Fig. 5.1b). Note that

$$\lim_{a\to 0} \delta_a(x) = 0 \qquad \text{for all } x \neq 0 ;$$

$$\int_{-\infty}^{\infty} \delta_a(x)\, dx = 1, \qquad \text{independent of } a ;$$

$$\lim_{a\to 0} \int_{-\infty}^{\infty} f(x)\delta_a(x)\, dx = f(0) .$$

The entire contribution to the integral, as $a \to 0$, comes from the neighborhood of $x = 0$. Therefore we may write symbolically,

$$\delta(x) = \lim_{a\to 0} \delta_a(x) = \lim_{a\to 0} \frac{1}{a\sqrt{\pi}} e^{-x^2/a^2}. \tag{5.20}$$

3. Another useful representation for the δ-function is

$$\delta(x) = \lim_{\epsilon \to 0} \delta_\epsilon(x) \equiv \lim_{\epsilon \to 0} \frac{1}{\pi} \frac{\epsilon}{x^2 + \epsilon^2}, \tag{5.21}$$

which the reader can establish as in the above example.

4. The final representation of the δ-function is slightly different from the preceding three. It will play a central role in the proof of Weierstrass's theorem, the basic result of this chapter. It is defined as

$$\delta_n(x) = \begin{cases} c_n(1 - x^2)^n & \text{for } 0 \leq |x| \leq 1, \\ 0 & \text{for } |x| > 1, \end{cases} \qquad n = 1, 2, 3, \cdots, \tag{5.22}$$

where the constant c_n must be determined so that

$$\int_{-1}^{1} \delta_n(x)\, dx = 1 . \tag{5.23}$$

The functions $\delta_n(x)$ form a sequence whose limit is a δ-function (see Fig. 5.1c). We shall show, first informally, then rigorously in the proof of Weierstrass's theorem, that

$$\lim_{n \to \infty} \int_{-1}^{1} f(x)\delta_n(x)\, dx = f(0) . \tag{5.24}$$

Thus

$$\lim_{n \to \infty} \delta_n(x) = \delta(x) . \tag{5.25}$$

This representation of the δ-function differs from the others in that the defining parameter n takes on integral values that increase to infinity, instead of decreasing continuously to zero.

First we determine the normalization constant c_n. We have

$$1/c_n = \int_{-1}^{1} (1 - x^2)^n\, dx = 2\int_0^1 (1 - x^2)^n\, dx . \tag{5.26}$$

Making the change of variable $x = \sin\theta$, we obtain

$$\frac{1}{c_n} = 2\int_0^{\pi/2} \cos^{2n+1}\theta\, d\theta = \frac{2^{n+1}n!}{1 \cdot 3 \cdot 5 \cdots (2n + 1)} . \tag{5.27}$$

From Eq. (5.27), it follows that

$$c_n = (2n + 1)!/2^{2n+1}(n!)^2 . \tag{5.28}$$

It is not clear from this expression for c_n how it behaves as $n \to \infty$. We can estimate its behavior from Eq. (5.26). Picking up again from there, we have

$$1/c_n = 2\int_0^1 (1 - x^2)^n\, dx \geq 2\int_0^{1/\sqrt{n}} (1 - x^2)^n\, dx , \tag{5.29}$$

since $1/\sqrt{n} \leq 1$ for all $n = 1, 2, \cdots$, and since the integrand is positive throughout $[0, 1]$. Now we shall show that for all n, the integrand $(1 - x^2)^n \geq 1 - nx^2$

for all x in $[0, 1]$. Consider the function $g(x) \equiv (1 - x^2)^n - (1 - nx^2)$. Since $g(0) = 0$, and

$$g'(x) = 2nx[1 - (1 - x^2)^{n-1}] > 0 \qquad \text{for all } 0 < x \leq 1 ,$$

$g(x)$ must be monotonically increasing in the interval $[0, 1]$. Therefore $g(x) \geq 0$, or $(1 - x^2)^n \geq (1 - nx^2)$, for all x in $[0, 1]$. Using this inequality in Eq. (5.29), we have

$$1/c_n \geq 2 \int_0^{1/\sqrt{n}} (1 - nx^2)\, dx = 4/3n^{1/2} > 1/n^{1/2} .$$

Therefore

$$c_n < n^{1/2} . \tag{5.30}$$

This result may also be obtained from Stirling's formula.

As $n \to \infty$, the contribution to the integral $\int_{-1}^{1} \delta_n(x)\, dx$ comes increasingly from the neighborhood surrounding the origin. To see this, note that for $0 < \delta < 1$,

$$\int_{-1}^{-\delta} \delta_n(x)\, dx = \int_{\delta}^{1} \delta_n(x)\, dx , \tag{5.31}$$

since $\delta_n(x)$ is an even function of x. Now

$$\int_{\delta}^{1} \delta_n(x)\, dx < n^{1/2}(1 - \delta^2)^n(1 - \delta) < n^{1/2}(1 - \delta^2)^n , \tag{5.32}$$

since $c_n < n^{1/2}$ [Eq. (5.30)], and $(1 - x^2)^n$ takes on its maximum value at $x = \delta$, and therefore $\int_{\delta}^{1}(1 - x^2)^n\, dx$ is bounded by the area of a rectangle of height $(1 - \delta^2)^n$ and base $(1 - \delta)$, since $0 < \delta < 1$. It is well known that the nth power of any positive number less than 1 will decrease more rapidly with n than any power of n will increase. In particular, the behavior of the term $(1 - \delta^2)^n$ will dominate the term $n^{1/2}$ as $n \to \infty$, and therefore

$$\lim_{n\to\infty} \int_{\delta}^{1} \delta_n(x)\, dx = 0 .$$

Since $\delta_n(x)$ is continuous and never negative, it follows that $\lim_{n\to\infty} \delta_n(x) = 0$ for $0 < x \leqq 1$. Since we have already arranged that $\int_{-1}^{1} \delta_n(x)\, dx = 1$ by our choice of c_n (Eq. 5.28), we obtain the result

$$\lim_{n\to\infty} \int_{-1}^{1} f(x)\delta_n(x)\, dx = f(0) . \tag{5.33}$$

5.4 WEIERSTRASS'S THEOREM: APPROXIMATION BY POLYNOMIALS

Weierstrass's famous approximation theorem proves that one can construct from the set of powers of x a sequence of polynomials which converges *uniformly* to any function that is continuous on the finite closed interval $[a, b]$. From this

result, we can prove that there exists a complete orthonormal set of polynomials on any interval $[a, b]$. Weierstrass's theorem is the starting point for proving the completeness properties of many of the functions of mathematical physics, such as the Legendre polynomials, the trigonometric functions (Fourier series), and the spherical harmonics.

Weierstrass's Theorem. If $f(x)$ is continuous on the closed interval $[a, b]$, there exists a sequence of polynomials $P_n(x)$ such that

$$\lim_{n\to\infty} P_n(x) = f(x)$$

uniformly on $[a, b]$.

Proof. We may assume without loss of generality that $f(x)$ is defined on $[0, 1]$. Suppose that $f(x)$ were defined on $[a, b]$. Then consider the function h defined by

$$h\left(\frac{x-a}{b-a}\right) \equiv f(x) .$$

Clearly, $f(a) = h(0)$, $f(b) = h(1)$, and any x in the interval $[a, b]$ corresponds to a z in $[0, 1]$. Thus if $h(z)$ can be approximated by polynomials in z, then since any polynomial in $z = (x-a)/(b-a)$ is also a polynomial in x, we can go from the polynomial approximating h to a polynomial approximating f. Furthermore, we can assume that $h(z)$ vanishes at $z = 0$ and $z = 1$, for if it does not, we define

$$g(z) = h(z) - h(0) - z[h(1) - h(0)]$$

for z in $[0, 1]$. Clearly, $g(0) = 0$ and $g(1) = 0$. Since $g(z)$ and $h(z)$ differ only by a polynomial, if we can approximate $g(z)$ by a polynomial, we can approximate $h(z)$ by that same polynomial plus the polynomial $h(0) + [h(1) - h(0)]z$.

Therefore we assume our original function $f(x)$ to be defined on $[0, 1]$ and to vanish at $x = 0$ and $x = 1$. We may define $f(x)$ as we choose outside $[0, 1]$, and we define it to be identically zero there.

Now we set

$$P_n(x) = \int_{-1}^{1} f(x+t)\delta_n(t)\,dt, \qquad 0 \le x \le 1 , \tag{5.34}$$

where $\delta_n(t)$ is the sequence of functions, defined in Eq. (5.22), that we claimed represented a δ-function:

$$\delta_n(t) = \begin{cases} c_n(1-t^2)^n & \text{for } 0 \le |t| \le 1 , \\ 0 & \text{for } |t| > 1 . \end{cases}$$

If that claim were true, then

$$\lim_{n\to\infty} P_n(x) = f(x) ,$$

and the proof would be complete. We mention this just to indicate the motivation behind the rigorous proof we now present. This proof will also demon-

strate that the sequence $\delta_n(t)$ as $n \to \infty$ really does have the properties that characterize a δ-function.

We are assuming that $f(x)$ vanishes outside $[0, 1]$. We may therefore write Eq. (5.34) as

$$P_n(x) = \int_{-x}^{1-x} f(x+t)\delta_n(t)\,dt\,,$$

where $f(x+t) \equiv 0$ whenever $t \leqq -x$ or $t \geqq 1-x$. By a simple change of variable $(t \to t - x)$, we obtain

$$P_n(x) = \int_0^1 f(t)\delta_n(t-x)\,dt = \int_0^1 f(t)c_n[1-(t-x)^2]^n\,dt\,.$$

This last integral shows clearly that $P_n(x)$ *is a polynomial* (of degree $2n$) in x. The coefficients of the powers of x are definite integrals over t. Thus $P_n(x)$ is a sequence of polynomials. We shall prove that this sequence converges uniformly to $f(x)$.

In analysis it is shown that a function which is continuous on a finite *closed* interval is uniformly continuous there. For those unfamiliar with this result, its meaning may be abstracted from the following example. The function $q(x) = x^{-1}$ is continuous on the *open* interval $(0, 1)$, but is not uniformly continuous there because

$$|q(x+\delta) - q(x)| = \frac{\delta}{x(x+\delta)}\,,$$

and this difference *cannot* be made arbitrarily small *for all* x by a *single* choice of δ. In fact, by choosing x sufficiently close to zero—yet still in $(0, 1)$—the difference can be made arbitrarily large for any given $\delta > 0$. Thus there does not exist a $\delta > 0$ *independent of* x such that $|q(x+\delta) - q(x)|$ is arbitrarily small, and the continuity is not uniform.

Since $f(x)$ is continuous on the closed interval $[0, 1]$, it is uniformly continuous there. As will be seen, *uniform* continuity is essential to this proof. Therefore we know that given any $\epsilon > 0$, there exists a δ such that

$$|f(x+\delta) - f(x)| < \epsilon$$

for all x in $[0, 1]$.

Now, using Eq. (5.34) for $P_n(x)$, we form the quantity

$$\begin{aligned} |P_n(x) - f(x)| &= \left|\int_{-1}^{1} [f(x+t) - f(x)]\delta_n(t)\,dt\right| \\ &\leq \int_{-1}^{1} |f(x+t) - f(x)|\delta_n(t)\,dt\,, \end{aligned}$$

since $\delta_n(t) \geq 0$ for all t in $[-1, 1]$. We now break up the range of integration into three parts:

$$\int_{-1}^{1} |f(x+t) - f(x)|\delta_n(t)\,dt = \int_{-1}^{-\delta} + \int_{-\delta}^{\delta} + \int_{\delta}^{1}.$$

Since $f(x)$ is continuous on a closed interval, it is bounded there. Let the maximum value of $|f(x)| = M$. Then both $\int_{-1}^{-\delta}$ and $\int_{\delta}^{1}$ are bounded by the quantity $2Mn^{1/2}(1-\delta^2)^n$. We may see this as follows:

$$\int_{\delta}^{1} |f(x+t) - f(x)|\delta_n(t)dt \leq \int_{\delta}^{1} |f(x+t)|\delta_n(t)\,dt + \int_{\delta}^{1} |f(x)|\delta_n(t)\,dt$$
$$\leq 2M\int_{\delta}^{1} \delta_n(t)\,dt < 2Mn^{1/2}(1-\delta^2)^n\,.$$

Here we have used the fact that $|f| < M$ and Eqs. (5.31) and (5.32).

We may estimate the remaining integral, $\int_{-\delta}^{\delta}$, by using the uniform continuity of $f(x)$, and interpreting the limits of integration as a δ which guarantees

$$|f(x+t) - f(x)| < \epsilon/2, \qquad \text{for } |t| < \delta.$$

We find

$$\int_{-\delta}^{\delta} |f(x+t) - f(x)|\delta_n(t)\,dt < \epsilon/2 \int_{-\delta}^{\delta} \delta_n(t)\,dt < \epsilon/2\,,$$

since $\int_{-\delta}^{\delta} \delta_n(t)\,dt < 1$.

Collecting these results, we have

$$|P_n(x) - f(x)| < 4Mn^{1/2}(1-\delta^2)^n + \epsilon/2\,.$$

The value of $n^{1/2}(1-\delta^2)^n$ for $0 < \delta < 1$ can be made arbitrarily small for large enough n and, in particular, smaller than $\epsilon/2$. Therefore there exists an N such that for $n > N$,

$$|P_n(x) - f(x)| < \epsilon\,,$$

for any arbitrarily small preassigned ϵ, that is,

$$\lim_{n\to\infty} |P_n(x) - f(x)| = 0.$$

This means that the sequence of polynomials $P_n(x)$ converges uniformly to the continuous function $f(x)$ on $[0, 1]$, and so the proof is complete. In fact, this holds for an arbitrary continuous function on an arbitrary finite closed interval $[a, b]$, as was demonstrated at the outset. QED

Thus Weierstrass's theorem tells us that there exists a set of coefficients

$$a_{nm}; \qquad n = 0, \cdots, \infty, \quad m = 0, \cdots, \mu = 2n\,,$$

such that $\sum_{m=0}^{m=\mu} a_{nm}\,x^m$ tends uniformly to $f(x)$ as $n \to \infty$. It most emphatically does not guarantee a uniformly convergent power series, which would consist of a set of coefficients a_m $(m = 0, \cdots, \infty)$ such that $\sum_{m=0}^{m=n} a_m x^m$ tends uniformly to $f(x)$ as $n \to \infty$. The point is that the Weierstrass coefficients a_{nm} are not independent of n for fixed m. As the approximation improves, by going to polynomials of higher degree, the earlier coefficients change. The theorem would be false if it claimed to produce a uniformly convergent power series. For example,

there is no *power* series that converges uniformly to the continuous function $\sqrt{x}$ in the interval [0, 1].

Weierstrass's theorem for approximation by a sequence of polynomials is in one sense, much stronger than Taylor's theorem for expansion in power series. To expand a function in a Taylor series, its derivatives of all orders must exist: it must be "analytic." In Weierstrass's theorem, only continuity is needed. Furthermore, Weierstrass's theorem demonstrates the existence of polynomial approximations outside the radius of convergence of a Taylor series. But there is, in general, no possibility of rearranging the uniformly convergent sequence of *polynomials* that approximate any continuous function so as to produce a convergent *power* series (Taylor expansion).

In Taylor expansions we need to know the function and its derivatives *locally*, at a point; the radius of convergence of the expansion may be finite or infinite. Weierstrass's theorem applies only to finite intervals, and we need to know the function, but not its derivatives, *globally*, over the entire interval.

As a by-product of this theorem, we have established rigorously that the representation of Dirac's delta function given in Eq. (5.22) has the properties claimed for it [Eq. (5.24)]. Thus Weierstrass, who proved this theorem in 1885 by the above method, anticipated Dirac considerably. In fact, he was not the only one. Heaviside used a closely related symbolic function before Dirac. Yet it was with Dirac's introduction of symbolic functions in his classic book, *The Principles of Quantum Mechanics*, which appeared in 1930, that their use became widespread. Only relatively recently was the theory of δ-functions, and other related symbolic functions, established rigorously by the mathematician Laurent Schwartz in his theory of distributions.

Weierstrass's theorem may also be extended to functions of several variables. By a straightforward generalization of the proof, it can be shown that if a function $f(x_1, x_2, \cdots, x_m)$ is continuous in each variable x_i for x_i in $[a_i, b_i]$ $(i = 1, 2, \cdots, m)$, f may be approximated uniformly by the polynomials

$$P_n(x_1, x_2, \cdots, x_m) = \int_{a_1}^{b_1}\int_{a_2}^{b_2}\cdots\int_{a_m}^{b_m} f(t_1, t_2, \cdots, t_m)\delta_n(t_1 - x_1)$$
$$\delta_n(t_2 - x_2)\cdots\delta_n(t_m - x_m)\, dt_1\, dt_2\cdots dt_m\,. \tag{5.35}$$

The completeness of the trigonometric functions follows from the special case $m = 2$, as we shall see in Section 5.6.

We close this section with the statement of an important consequence of our basic result. From Section 5.2 it follows that if the set P_n defined in the proof of Weierstrass's theorem approximate any continuous function f uniformly, they also approximate any continuous function *in the mean*; that is, given any ϵ, there exists an n such that $||f - P_n|| < \epsilon$. Now it can be shown that any function in the Hilbert space of square-integrable functions can be approximated arbitrarily closely in the mean by a continuous function. This is discussed further in Section 8.7. It follows that *any* function in this Hilbert space can be approximated *in the mean* by some P_n. This can be seen as follows.

Let ϕ be a function in Hilbert space. Then the difference $\phi - P_n$ can be written as

$$\phi - P_n = (\phi - f) + (f - P_n) ,$$

where f is a continuous function. By the triangle inequality, we have

$$||\phi - P_n|| \leq ||\phi - f|| + ||f - P_n|| .$$

Now f can be chosen so that $||\phi - f||$ is as small as we please; and by Weierstrass's theorem, n can be chosen so that $||f - P_n||$ is as small as we please. Thus P_n approximates ϕ, an arbitrary function in Hilbert space, arbitrarily closely in the mean. This fact will be needed in what follows. A complete proof can be found in most analysis texts, e.g., Riesz and Nagy (Section 46) or Rudin (Theorem 10.38).

5.5 LEGENDRE POLYNOMIALS

We are now in a position to demonstrate that there exists a *complete* orthonormal set of polynomials on the finite closed interval $[a, b]$. The proof of completeness follows from Weierstrass's theorem. After establishing the general result, we shall examine in detail one very important special case—the complete orthonormal set of polynomials on the interval $[-1, 1]$, which, apart from constant normalization factors, is the set of Legendre polynomials. We shall construct the first three functions in this set explicitly, and then provide a formula for calculating all the others.

Weierstrass's theorem tells us that any continuous function f on $[a, b]$ can be approximated uniformly by a sequence of polynomials:

$$P_n(x) = \sum_{m=0}^{2n} a_{nm} x^m .$$

Since the set $\{x^n, n = 0, 1, \cdots\}$ is linearly independent, we may apply the Gram-Schmidt orthogonalization process to construct from it an orthonormal basis $\{Q_n(x)\}$, where $Q_n(x)$ is a polynomial of degree n. We may now express the original functions as finite linear combinations of the orthonormal set:

$$x^m = \sum_{i=0}^{m} c_{mi} Q_i(x) .$$

It follows that the polynomials P_n, which approximate the function f uniformly, may be expressed as

$$P_n(x) = \sum_{m=0}^{2n} a_{nm} \sum_{i=0}^{m} c_{mi} Q_i(x) .$$

We now prove that the orthonormal set $\{Q_n\}$ is complete by showing that it is closed (Theorem 5.2). The orthonormal set $\{Q_n\}$ will be closed if the only function orthogonal to all the Q_n is the zero function. Assume that we have a function in Hilbert space such that $(f, Q_n) = 0$ for all n. It follows immediately from the above equation that $(f, P_n) = 0$ for all n, since the P_n are linear combinations of the Q_n. However, we know from Weierstrass's theorem that f may be approximated in the mean by the P_n. Thus for any ϵ there exists an n such that

$||f - P_n|| < \epsilon$. But since $(f, P_n) = 0$,

$$||f - P_n|| \equiv (f - P_n, f - P_n)^{1/2} = [||f||^2 + ||P_n||^2]^{1/2} .$$

Therefore

$$||f||^2 + ||P_n||^2 < \epsilon^2 .$$

But this implies that $||f||$ is arbitrarily small, and hence f is equal to zero (almost everywhere). Therefore the orthonormal set $\{Q_n\}$ is closed and hence, by Theorem 5.2, it is a complete orthonormal set of polynomials on $[a, b]$.

The completeness of the set Q_i means (according to Definition 5.6) that there exists a set of of constants $\{a_i\}$ such that any function g in the Hilbert space can be approximated in the mean by the sequence of partial sums

$$g_n = \sum_{i=0}^{n} a_i Q_i(x) .$$

But the a_i are independent of n. Thus as we extend the sum to infinity, the approximation improves *without the earlier a_i changing*. Therefore we may say that there exists an infinite *series*

$$\sum_{i=0}^{\infty} a_i Q_i$$

which approximates g in the mean. Symbolically, we write

$$g(x) \doteq \sum_{i=0}^{\infty} a_i Q_i(x) .$$

We have shown that if we have a complete orthonormal set of functions, the coefficients which provide the best approximation in the mean are the Fourier coefficients. Thus the expansion coefficients in the infinite series are given by

$$a_i = (Q_i, g) .$$

Let us now apply these considerations to a concrete case. We shall use the Gram-Schmidt orthogonalization process applied to the basis $\{1, x, x^2, \cdots\}$ to construct the set of orthonormal polynomials on the interval $[-1, 1]$. By the previous result we know that this set will be complete.

The first member of the orthonormal set is $\bar{P}_0(x) = (\frac{1}{2})^{1/2}$, which is evidently normalized for x in $[-1, 1]$. Using the prescription of Section 4.3, we have

$$\bar{P}_0(x) = (\tfrac{1}{2})^{1/2} ,$$

$$\bar{P}_1 = \frac{x - 1/2^{1/2}\int_{-1}^{1} 1/2^{1/2}x\,dx}{\left\|x - 1/2^{1/2}\int_{-1}^{1} 1/2^{1/2}x\,dx\right\|} = \left(\frac{3}{2}\right)^{1/2} x ,$$

$$\bar{P}_2 = \frac{x^2 - 1/2^{1/2}\int_{-1}^{1} 1/2^{1/2}x^2\,dx - (\frac{3}{2})^{1/2}x\int_{-1}^{1} (\frac{3}{2})^{1/2}x^3\,dx}{\left\|x^2 - 1/2^{1/2}\int_{-1}^{1} 1/2^{1/2}x^2\,dx - (\frac{3}{2})^{1/2}x\int_{-1}^{1} (\frac{3}{2})^{1/2}x^3\,dx\right\|} = \left(\frac{5}{2}\right)^{1/2}\left(\frac{3}{2}x^2 - \frac{1}{2}\right). \tag{5.36}$$

This is a tedious process; we have carried it out for these three cases only to show that one can construct orthonormal functions directly. One would hope that there is a better way to compute the orthonormal polynomials of higher order, and, indeed, there is.

We claim that a general formula for the complete orthonormal set of polynomials on $[-1, 1]$ is

$$\bar{P}_n(x) = \left(\frac{2n+1}{2}\right)^{1/2} \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n . \tag{5.37}$$

The orthogonal, but unnormalized, polynomials

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n \tag{5.38}$$

are known as the *Legendre polynomials*. Equation (5.38) is called *Rodrigues' formula*.

Thus the Legendre polynomials differ from the orthonormal set $\{\bar{P}_n(x)\}$ by constant multiplicative factors. The problem now is to show that the Rodrigues formula gives the *same* set of orthonormal polynomials as the Gram-Schmidt orthogonalization process. Direct computation verifies that they agree through the polynomials of second degree. To show that the Rodrigues formula holds for all orders, we must show that:

1. On any fixed interval there can be only one complete orthonormal set of polynomials in which the nth polynomial is of degree n.
2. The $\bar{P}_n(x)$ given by the Rodrigues formula do in fact form a complete orthonormal set on $[-1, 1]$.

The uniqueness property of the complete orthonormal set of polynomials is embodied in the Gram-Schmidt process, which can generate one and only one complete orthonormal set of polynomials (up to a phase factor) on $[-1, 1]$, or, in fact, on any given interval. Hence we have, in fact, already proved condition (1).

To demonstrate condition (2), we now show that the set of polynomials generated by the Rodrigues formula for the interval $[-1, 1]$ is orthonormal, and that the function $P_n(x)$ given by Rodrigues' formula is a polynomial of degree n. Then completeness follows by our earlier proof. It is immediately clear that $P_n(x)$ is a polynomial of degree n. We prove orthonormality in the following theorem.

Theorem 5.3.

$$(\bar{P}_n, \bar{P}_m) = \int_{-1}^{1} \bar{P}_n(x)\bar{P}_m(x)\, dx = \delta_{nm} . \tag{5.39}$$

Proof. We first prove orthogonality. We denote d^n/dx^n by d^n, and suppose that $n > m$. Dropping constant factors, where we have integrated by parts,

we have

$$\int_{-1}^{1} \bar{P}_n \bar{P}_m \, dx = \int_{-1}^{1} [d^n(x^2 - 1)^n] [d^m(x^2 - 1)^m] \, dx$$
$$= [d^{n-1}(x^2 - 1)^n] [d^m(x^2 - 1)^m] \Big]_{-1}^{1}$$
$$- \int_{-1}^{1} [d^{n-1}(x^2 - 1)^n] [d^{m+1}(x^2 - 1)^m \, dx] ,$$

Since

$$d^{n-1}(x^2 - 1)^n = (\text{a polynomial}) \cdot (x^2 - 1),$$

the first term vanishes upon putting in the limits ± 1, leaving the second term. Therefore, after n partial integrations, we have

$$\int_{-1}^{1} (-1)^n (x^2 - 1)^n d^{m+n} (x^2 - 1)^m \, dx ,$$

since the term which is evaluated at ± 1 always vanishes because it is proportional to some power of $(x^2 - 1)$. Now, since $n > m$, $n + m > 2m$ and so $d^{n+m}(x^2 - 1)^m = 0$. Therefore

$$\int_{-1}^{1} \bar{P}_n \bar{P}_m \, dx = 0 \qquad \text{for } m \neq n .$$

If $m = n$, then as before (but putting in constant factors),

$$\int_{-1}^{1} \bar{P}_n \bar{P}_n \, dx = \frac{(2n + 1)(-1)^n}{2^{2n+1}(n\,!)^2} \int_{-1}^{1} (x^2 - 1)^n d^{2n} (x^2 - 1)^n \, dx .$$

But $(x^2 - 1)^n$ is a polynomial of degree $2n$, so the $(2n)^{\text{th}}$ derivative is just $(2n)!$. The integral becomes

$$\frac{(-1)^n (2n + 1)!}{2^{2n+1}(n!)^2} \int_{-1}^{1} (x^2 - 1)^n \, dx = 1 ,$$

where we have used Eqs. (5.26) and (5.28).

The functions $\bar{P}_n(x) = [(2n + 1)^{1/2}/2] \, P_n(x)$ are therefore a complete orthonormal set on the interval $[-1, 1]$, and they are the only such set.

We shall now work out several other results pertaining to the Legendre polynomials $P_n(x)$. Our approach will be inductive and straightforward, a continuation of the techniques used to prove orthonormality. There are more elegant ways to obtain most of these results which will be used later to deal with both the Legendre polynomials and other complete sets of functions. But for the moment we proceed in a more pedestrian manner.

First we shall derive from the Rodrigues formula an explicit expression for $P_n(x)$ as a polynomial. We apply the binomial theorem to the factor $(x^2 - 1)^n$ to obtain

$$(x^2 - 1)^n = \sum_{m=0}^{n} \binom{n}{m} x^{2m} (-1)^{n-m} ,$$

where $\binom{n}{m}$ is the binomial coefficient:

$$\binom{n}{m} \equiv \frac{n!}{m!(n-m)!} . \tag{5.40}$$

Then, taking the nth derivative, we obtain

$$P_n(x) = \frac{1}{2^n n!} d^n (x^2 - 1)^n = \frac{1}{2^n n!} \sum_{m \geq p}^{n} (-1)^{n-m} \binom{n}{m} \frac{(2m)!}{(2m-n)!} x^{2m-n} , \tag{5.41}$$

where $p = n/2$ if n is even and $p = (n+1)/2$ if n is odd.

We next prove that $P_n(1) = 1$ for all n. In itself this is not of staggering importance, although it is interesting. We prove it now to illustrate an important technique. (We shall have an easier way later.) We write

$$P_n(x) = \frac{1}{2^n n!} d^n[(x-1)^n (x+1)^n] ,$$

and evaluate the nth derivative of the product by Leibnitz's rule (the "binomial expansion" for the nth derivative of a product):

$$d^n(uv) = \sum_{m=0}^{n} \binom{n}{m} d^m u d^{n-m} v = u d^n v + n d u d^{n-1} v + \frac{n(n-1)}{2!} d^2 u d^{n-2} v + \cdots$$
$$+ \frac{n!}{k!(n-k)!} d^k u d^{n-k} v + \cdots + v d^n u. \tag{5.42}$$

We now take $u = (x-1)^n$ and $v = (x+1)^n$. It is clear that the only nonzero term in the Leibnitz expansion at $x = 1$ is the term $m = n$, in which the factor $u = (x-1)^n$ is "differentiated down" to the constant $n!$ and hence does not vanish at $x = 1$. Therefore

$$d^n[(x-1)^n (x+1)^n]\Big|_{x=1} = d^n (x-1)^n d^o (x+1)^n \Big|_{x=1} = n! 2^n ,$$

and consequently,

$$P_n(1) = 1 \qquad \text{for all } n . \tag{5.43}$$

The Legendre polynomials arose originally as the solution to a differential equation, now called Legendre's equation:

$$(1 - x^2) P_n''(x) - 2x P_n'(x) + n(n+1) P_n(x) = 0 . \tag{5.44}$$

To prove that the elements of the complete orthonormal set of polynomials on $[-1, 1]$ satisfy this equation, we begin with the identity

$$(x^2 - 1) \frac{d}{dx} (x^2 - 1)^n = 2nx(x^2 - 1)^n .$$

Now differentiate both sides of this identity $n + 1$ times. With the help of

Leibnitz's formula (Eq. 5.42), we obtain for the left-hand side

$$d^{n+1}[(x^2-1)d(x^2-1)^n] = (x^2-1)d^{n+2}(x^2-1)^n + 2(n+1)xd^{n+1}(x^2-1)^n + n(n+1)d^n(x^2-1)^n ,$$

and for the right-hand side, we have

$$d^{n+1}[2nx(x^2-1)^n] = 2nx\, d^{n+1}(x^2-1)^n + 2n(n+1)d^n(x^2-1)^n .$$

The difference of these two equations must vanish. Therefore

$$(x^2-1)d^{n+2}(x^2-1)^n + 2x\, d^{n+1}(x^2-1)^n - n(n+1)d^n(x^2-1)^n = 0 .$$

Now, using the Rodrigues formula for $P_n(x)$, we obtain

$$(x^2-1)P_n'' + 2x\, P_n' - n(n+1)P_n = 0 , \tag{5.45}$$

which is equivalent to Legendre's equation (5.44). It may also be written in the form

$$[(1-x^2)P_n'(x)]' + n(n+1)P_n(x) = 0 . \tag{5.46}$$

Thus the $P_n(x)$ are solutions of Legendre's equations; but they are not the only solutions. There exists another linearly independent solution called, collectively, the Legendre *functions* of the second kind, denoted by $Q_n(x)$. However, these functions are not finite at $x = \pm 1$ as are the $P_n(x)$, and so are excluded as possible solutions in many physical applications; they are not polynomials.

There is an alternative proof of the orthogonality of the Legendre polynomials which follows directly from the differential equation (5.46) for $P_n(x)$. We know that P_n and P_m $(n \neq m)$ satisfy the equations

$$\frac{d}{dx}[(1-x^2)P_n'] + n(n+1)P_n = 0 ,$$

$$\frac{d}{dx}[(1-x^2)P_m'] + m(m+1)P_m = 0 .$$

Multiplying these equations by P_m and P_n respectively, subtracting them, and integrating from -1 to $+1$, we obtain

$$\begin{aligned}
&(n-m)(n+m+1)\int_{-1}^{1} P_nP_m\, dx \\
&= \int_{-1}^{1}\left[P_n\frac{d}{dx}((1-x^2)P_m') - P_m\frac{d}{dx}((1-x^2)P_n')\right]dx \\
&= \left\{\Big[P_n(1-x^2)P_m'\Big]_{-1}^{1} - \int_{-1}^{1} dx\,(1-x^2)P_m'P_n' \right. \\
&\qquad \left. - \Big[P_m(1-x^2)P_n'\Big]_{-1}^{1} + \int_{-1}^{1} dx\,(1-x^2)P_n'P_m'\right\} = 0 ;
\end{aligned}$$

here the two integrals cancel and the two other terms vanish. Since $n \neq m$, it

follows that

$$(P_n, P_m) \equiv \int_{-1}^{1} P_n P_m \, dx = 0 .$$

This is actually a special case of a proof that holds for a wide variety of functions arising as solutions to the Sturm-Liouville problem, which we shall discuss in Section 5.10. The proof depends on the fact that the differential equations obeyed by orthogonal functions have a certain form.

We have sampled only a few of the properties of the complete orthonormal set of polynomials on $[-1, 1]$. We shall encounter them many times in the course of the book (we met them once before in the multipole expansion (Eqs. 1.100, 1.101, 1.102), and shall discuss their properties in more detail. But our main concern for the moment is the *completeness* of this set of polynomials, and other sets of functions—not all the special properties these sets of functions happen to have.

We now turn to the single most important complete set of functions, the ancestor of them all: the trigonometric functions and Fourier series.

5.6 FOURIER SERIES

The completeness properties of the set of trigonometric functions $\{\sin n\theta, \cos n\theta, n = 0, 1, 2, \cdots, \infty\}$ can be deduced from Weierstrass's theorem for two variables. Equation 5.35 tells us that any function $g(x, y)$ which is continuous in both variables on specified finite closed intervals may be approximated uniformly by the the sequence of functions

$$g_N(x, y) = \sum_{n,m=0}^{N} A_{nm}^{(N)} x^n y^m ;$$

that is, $\lim_{N\to\infty} g_N(x, y) = g(x, y)$, uniformly in x and y. The coefficients $A_{nm}^{(N)}$ are not independent of N for fixed n and m, so this is *not* a power-series expansion.

If we change to polar coordinates and restrict the domain of definition to the unit circle, then $x = \cos\theta$ and $y = \sin\theta$ so that

$$g(\cos\theta, \sin\theta) = f(\theta) = \lim_{N\to\infty} \sum_{n,m=0}^{N} A_{nm}^{(N)} \cos^n\theta \sin^m\theta . \tag{5.47}$$

Clearly, the only functions $f(\theta)$ which can satisfy this last equation are periodic functions with periodicity 2π; this is a consequence of restricting x and y to the unit circle. We shall generalize later to include functions having periodicities different from 2π.

Using Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta ,$$

we obtain expressions for the nth powers of $\sin\theta$ and $\cos\theta$:

$$\cos^n\theta = \left[\frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)\right]^n , \qquad \sin^n\theta = \left[\frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)\right]^n .$$

Then Eq. (5.47) can be rewritten in the form

$$f(x) = \lim_{M\to\infty} f_M(x) \equiv \lim_{M\to\infty} \sum_{n=-M}^{M} \frac{c_n^{(M)}}{(2\pi)^{1/2}} e^{inx}, \tag{5.48a}$$

where we have inserted the factor $(2\pi)^{1/2}$ for later convenience, and have replaced the variable θ by x to emphasize the general nature of the result. Clearly, we may write equivalently

$$f(x) = \lim_{M\to\infty} f_M(x) = \lim_{M\to\infty} \left[\frac{a_0^{(M)}}{2} + \sum_{n=1}^{M} (a_n^{(M)} \cos nx + b_n^{(M)} \sin nx) \right]. \tag{5.48b}$$

The complex expansion coefficients in the exponential form of the series, and the real coefficients in the trigonometric form, are related as follows:

$$\begin{aligned} a_0^{(M)} &= \left(\frac{2}{\pi}\right)^{1/2} c_0^{(M)}, & c_0^{(M)} &= \left(\frac{\pi}{2}\right)^{1/2} a_0^{(M)}, \\ a_n^{(M)} &= \frac{1}{(2\pi)^{1/2}} (c_n^{(M)} + c_{-n}^{(M)}), \quad \text{or} & c_n^{(M)} &= \left(\frac{\pi}{2}\right)^{1/2} (a_n^{(M)} - ib_n^{(M)}), \\ b_n^{(M)} &= \frac{i}{(2\pi)^{1/2}} (c_n^{(M)} - c_{-n}^{(M)}), & c_{-n}^{(M)} &= \left(\frac{\pi}{2}\right)^{1/2} (a_n^{(M)} + ib_n^{(M)}). \end{aligned} \tag{5.49}$$

The superscript M reminds us that the coefficients in the sequence are *not* independent of M; the coefficients of earlier terms in the sequence may change as M increases.

Thus any continuous function $f(x)$, for which $f(x) = f(x + 2\pi)$, can be approximated uniformly by a sequence of trigonometric polynomials:

$$f_M(x) = \sum_{n=-M}^{M} \frac{c_n^{(M)}}{(2\pi)^{1/2}} e^{inx} = \frac{a_0^{(M)}}{2} + \sum_{n=1}^{M} (a_n^{(M)} \cos nx + b_n^{(M)} \sin nx).$$

Now suppose that $f(x)$ is any continuous function on the closed interval $[a, a + 2\pi]$, and assume further that $f(x)$ does not satisfy the periodicity condition $f(a) = f(a + 2\pi)$ at the endpoints of the interval. It is clear that for any such function f, we can find a continuous function $g(x)$ that both satisfies the periodicity requirement and is such that the quantity

$$\int_a^{a+2\pi} |f(x) - g(x)|^2 \, dx$$

can be made arbitrarily small. (See Courant-Hilbert for proof.) Therefore the trigonometric sequence that converges uniformly to $g(x)$ will converge *in the mean* to $f(x)$ on the interval $[a, a + 2\pi]$. In fact, since any function in Hilbert space can be approximated in the mean to arbitrary precision by some continuous function (see Section 5.4), this same result holds for any function in Hilbert space.

We saw in Section 5.1 that the set of functions

$$\left\{ \frac{1}{(2\pi)^{1/2}} e^{inx}, \; n = 0, \pm 1, \cdots \right\}$$

is orthonormal on the interval $[-\pi, \pi]$. We shall denote this orthonormal set of functions by $\{T_n\}$. We shall now show that this set is a *complete* orthonormal set. We do this, exactly as we did for the orthonormal set of polynomials $\{Q_n\}$ in the last section, by proving that the orthonormal set is closed. Thus, assuming that $(f, T_n) = 0$ for all n, we want to prove $f = 0$. It follows from Eq. (5.48a) that $(f, f_M) = 0$ for all M, because f_M is just a linear combination of the orthonormal funtions T_n. However, we know that any function f may be approximated in the mean by the sequence f_M; that is,

$$||f - f_M|| = [||f||^2 + ||f_M||^2]^{1/2} < \epsilon .$$

Thus $f = 0$ (almost everywhere). Therefore the orthonormal set $\{T_n\}$ is closed, and hence by Theorem 5.2, it is a complete orthonormal set on $[-\pi, \pi]$. Clearly, the set of trigonometric functions which is orthonormal on $[-\pi, \pi]$,

$$\left\{\frac{1}{(2\pi)^{1/2}}, \qquad \frac{1}{\pi^{1/2}} \sin nx, \qquad \frac{1}{\pi^{1/2}} \cos nx, \qquad n = 1, 2, \cdots\right\},$$

is also complete.

It follows from the completeness of the orthonormal set $\{T_n\}$ that we may approximate an arbitrary function f in the mean by an infinite series of the T_n. We write symbolically,

$$f(x) \doteq \sum_{n=-\infty}^{\infty} c_n T_n(x) = \sum_{n=-\infty}^{\infty} \frac{c_n}{(2\pi)^{1/2}} e^{inx} = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx). \quad (5.50)$$

The expansion coefficients which here are constants, independent of M (compare Eq. 5.48a), are given by

$$c_n = (T_n, f) = \frac{1}{(2\pi)^{1/2}} \int_{-\pi}^{\pi} f(x) e^{-inx}\, dx . \quad (5.51)$$

The coefficients a_n and b_n of the trigonometric series may be computed using Eqs. (5.49):

$$a_n = \frac{1}{(2\pi)^{1/2}} (c_n + c_{-n}) = \left(\frac{1}{(2\pi)^{1/2}}\right)^2 \int_{-\pi}^{\pi} f(x)(e^{-inx} + e^{inx})\, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx \qquad \text{for } n = 0, 1, \cdots \quad (5\ 52)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx \qquad \text{for } n = 1, 2, \cdots$$

The series (5.50) with the coefficients (5.52) is known as the *Fourier series.*

Since the functions

$$T_n(x) = \frac{1}{(2\pi)^{1/2}} e^{inx}, \qquad n = 0, \pm 1 \cdots$$

form a complete orthonormal set, the completeness relation (Eq. 5.10) must